

CHAPTER  
23

# Pathogens and Infection

Infectious diseases currently cause about one-quarter of all human deaths worldwide, more than all forms of cancer combined and second only to cardiovascular diseases. There is a continuing heavy burden of ancient diseases such as tuberculosis and malaria, and these are increasingly difficult to treat because of rising drug resistance. Newer infectious diseases continually emerge, including the current pandemic (worldwide epidemic) of COVID-19 (coronavirus disease 2019), which was first clinically observed in 2019 and has since infected hundreds of millions of people and caused millions of deaths worldwide. Moreover, some diseases long thought to result from other causes are now recognized to be associated with infections. Most gastric ulcers, for example, are caused not by stress or spicy food, but by infection of the stomach lining by the bacterium *Helicobacter pylori*.

The burden of infectious diseases is not spread equally across the planet. Poorer countries and communities suffer disproportionately, often due to poor public sanitation and overburdened health systems, as occurred in Haiti in 2010 with a severe cholera outbreak after a devastating earthquake. Some infectious diseases, however, occur primarily or exclusively in industrialized communities: Legionnaires' disease, for example, a bacterial infection of the lungs, commonly spreads through air-conditioning systems.

Since the mid-1800s, physicians and scientists have struggled to identify the microbes—collectively called **pathogens**—that are capable of causing infectious diseases. More recently, the advent of microbial genetics and molecular cell biology has greatly enhanced our understanding of the causes and mechanisms of infectious diseases. We now know that pathogens frequently exploit the attributes of their host's cells in order to infect them. This understanding can give us new insights into normal cell biology, as well as strategies for treating and preventing infections.

Although pathogens are understandably a focus of attention, only a relatively small fraction of the microbial species we encounter are pathogens. Much of the biomass of Earth is made up of microbes. They produce everything from the oxygen we breathe to the soil nutrients we use to grow food and represent the bottom of the ecological pyramid. Even those species of microbes that colonize the human body do not generally cause disease. Many of these microbes have a beneficial effect on the health of the organism, assisting its normal development and physiology.

In this chapter, we give an overview of the different kinds of pathogens. We then discuss the cell biology of infection—the molecular interactions between pathogens and their hosts. Finally, we discuss an emerging concept that the many microbes that normally colonize our body, the so-called human **microbiota**, may actually benefit our health. In Chapter 24, we consider how our innate and adaptive immune systems collaborate to defend us against pathogens while leaving our microbiota intact.

## IN THIS CHAPTER

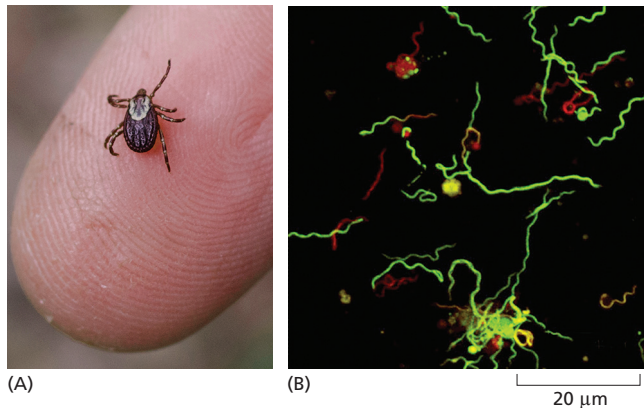
Introduction to Pathogens

Cell Biology of Pathogen Infection

The Human Microbiota

## INTRODUCTION TO PATHOGENS

We normally think of pathogens as hostile invaders, but a pathogen, like any other organism, is simply exploiting an available niche in which to live and procreate. Living on or in a host organism is a very effective strategy, and it is possible that



**Figure 23-1** Parasitism at many levels.

(A) Most animals harbor parasites, an example being the blacklegged tick, or deer tick (*Ixodes scapularis*), shown here on a human finger. Although ticks of this species thrive on white-tailed deer and other wild mammals, they can also live on humans. (B) Ticks themselves harbor their own parasites including the bacterium *Borrelia burgdorferi*, stained here with a vital dye that labels living bacteria green and dead bacteria red. These long spiral-shaped bacteria live in deer ticks and can be transmitted to humans during a tick's blood meal. *B. burgdorferi* causes Lyme disease, which is characterized by a bull's-eye-shaped skin rash and fever; if the infection is left untreated, various complications can result, including arthritis and neurological abnormalities. The idea that parasites have their own parasites was noted by Jonathan Swift in 1733: "So, naturalists observe, a flea Has smaller fleas that on him prey; And these have smaller still to bite 'em; And so proceed ad infinitum." (A, National Geographic Image Collection/Alamy Stock Photo; B, courtesy of M. Embers.)

every organism on Earth is subject to some type of infection (**Figure 23-1**). Even bacteria can be infected by viruses called bacteriophages (see **Figure 1-37**). It is not surprising then that many microorganisms have evolved the ability to survive and reproduce in the human body, a nutrient-rich, warm, and moist environment that remains at a uniform temperature and constantly renews itself. In this section, we discuss some of the common features that microorganisms must have in order to colonize the human body or cause disease, and we explore the wide variety of organisms that are known to cause disease.

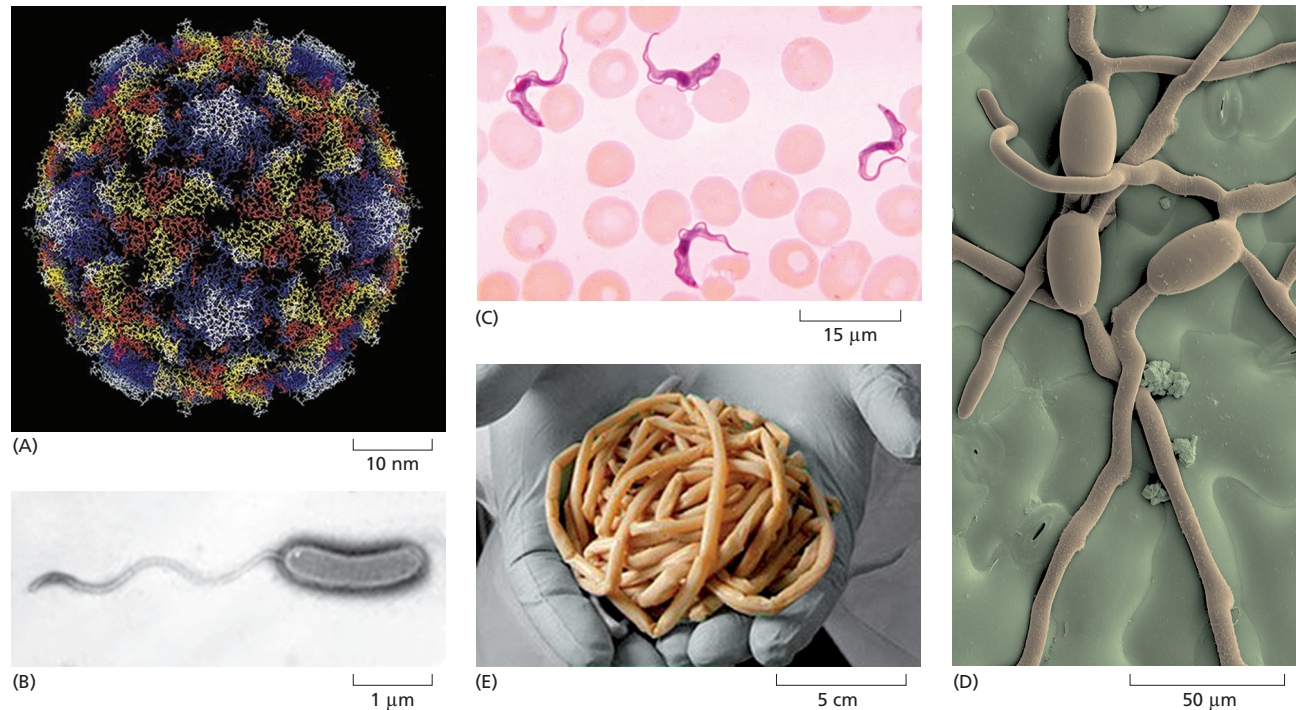
### Pathogens Can Be Viruses, Bacteria, or Eukaryotes

Many types of microbes cause disease in humans, and others reside on or in our bodies without causing harm. The most familiar are viruses and bacteria. Viruses cause diseases ranging from COVID-19 and Ebola virus disease to the common cold. Viruses are essentially fragments of nucleic acid (DNA or RNA) that generally encode a relatively small number of gene products and are wrapped in a protective shell of proteins (**Figure 23-2A**) and (in some cases) an outer membrane envelope. Much larger and more complex than viruses, bacteria are prokaryotic cells that perform most of their basic metabolic functions themselves, relying on the host primarily for nutrition (**Figure 23-2B**). Bacteria cause illnesses ranging from tuberculosis, which causes roughly 1.4 million deaths per year, to pneumonia, as well as diarrheal and sexually transmitted diseases.

Some other infectious agents are eukaryotic organisms. These range from single-celled protozoa (**Figure 23-2C**) and fungi (**Figure 23-2D**) to large, complex metazoa such as parasitic worms. One of the most common human parasites, shared by about a billion people at present, is the nematode worm *Ascaris lumbricoides*, which infects the gut (**Figure 23-2E**). It resembles its harmless nematode cousin *Caenorhabditis elegans*, which is used as a model organism for genetic and developmental biological research (see **Figure 1-42**). *C. elegans*, however, is only about 1 mm in length, whereas *Ascaris* can reach 30 cm.

### Pathogens Interact with Their Hosts in Different Ways

Although the ability of a particular microorganism to cause disease depends on many factors, it requires that the pathogen possess specialized pathogenic characteristics that allow it to live in humans. One way we distinguish between pathogens is by classifying them as primary versus opportunistic on the basis of the circumstances under which they cause human disease. **Primary pathogens** can cause overt disease in most healthy, nonimmune people. **Opportunistic pathogens** do not cause disease in healthy people but can cause illness in individuals suffering from other conditions. Some primary pathogens cause acute, life-threatening epidemic infections and spread rapidly from one sick or dying host to another; historically important examples include the bacterium *Vibrio cholerae*, which causes cholera, and the sudden acute respiratory syndrome coronavirus 2 (SARS-CoV-2) and influenza viruses, which cause COVID-19



and flu, respectively. Others may persistently infect a single individual for years without causing overt disease; examples include the bacterium *Mycobacterium tuberculosis*, which causes tuberculosis, and the intestinal worm *Ascaris* (see Figure 23-2E). Although these potential primary pathogens can make some people critically ill, billions of people carry these foreign organisms in an asymptomatic way, often unaware that they are infected.

In order to survive and multiply, a successful pathogen must be able to (1) enter the host (usually by breaking an epithelial barrier); (2) find a nutritionally compatible niche in the host's body; (3) avoid, subvert, or circumvent the host's innate and adaptive immune responses; (4) replicate, using host resources; and (5) exit one host and spread to another. Pathogens have evolved various mechanisms that exploit the biology of their host organisms to help accomplish these five tasks. For some pathogens, these mechanisms are adapted to a unique host species, whereas for others the mechanisms are sufficiently general to permit invasion, survival, and replication in a wide variety of hosts. Because pathogens have evolved the ability to interface directly with the molecular machinery of host cells, we have learned a great deal about cell biological principles by studying them.

Our constant exposure to pathogens has strongly influenced human evolution. In modern times, humans have learned how to limit the ability of pathogens to infect us through improvements in public health measures and childhood nutrition, vaccines, antimicrobial drugs, and routine testing of blood used for transfusions. As we learn more about the mechanisms by which pathogens cause disease (called *pathogenesis*), we will devise new ways to supplement or augment our immune systems in fighting infections.

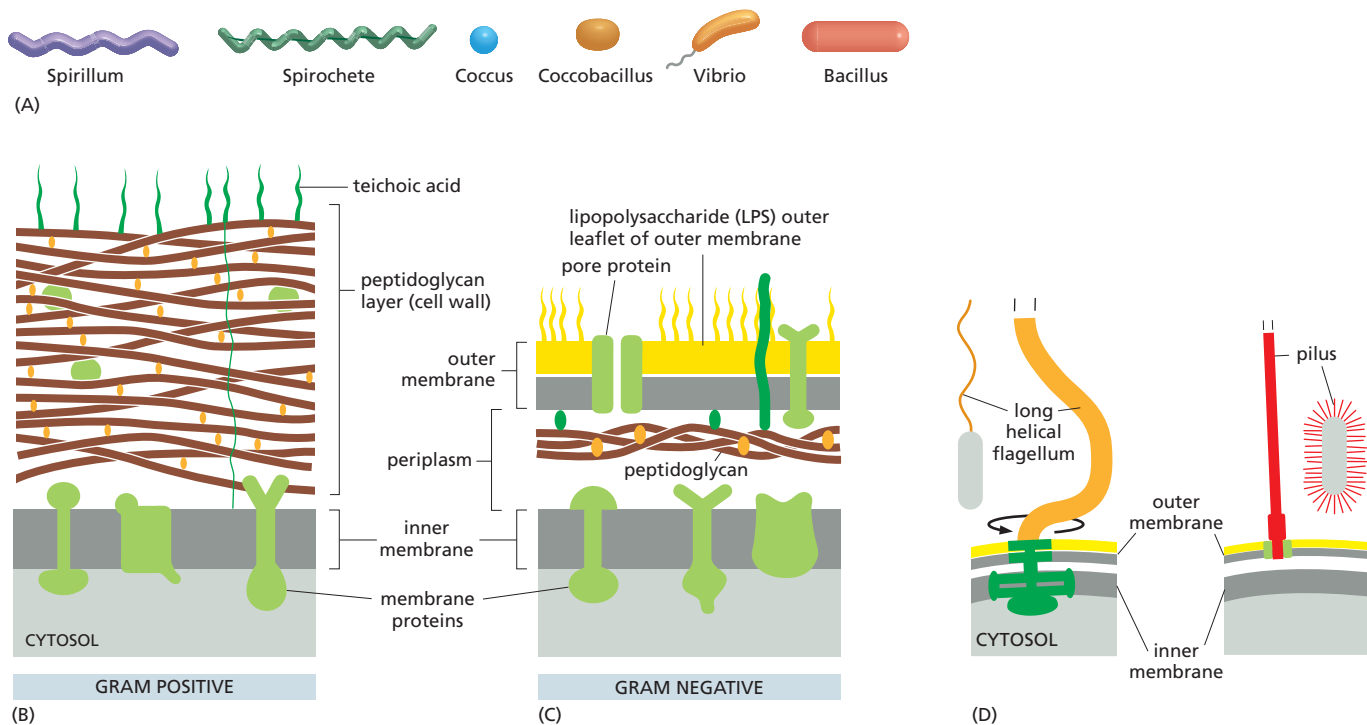
We now introduce the basic features of each of the three major types of pathogens—bacteria, eukaryotic parasites, and viruses—before we examine the detailed mechanisms used by these pathogens to infect their hosts.

### Bacteria Are Diverse and Occupy a Remarkable Variety of Ecological Niches

**Bacteria** are small but highly sophisticated cells whose organization and behaviors have attracted the attention of many scientists for well over a century. Bacteria are classified broadly on the basis of their cellular and molecular features. One

**Figure 23-2 Pathogens in many forms.**

(A) The structure of the protein coat, or capsid, of poliovirus. This virus was once a common cause of paralysis, but the disease (poliomyelitis) has been greatly reduced by widespread vaccination. (B) The bacterium *Vibrio cholerae*, the causative agent of the epidemic diarrheal disease cholera, shown here with its attached flagellum. (C) The protozoan parasite *Trypanosoma brucei* (purple) in a field of erythrocytes (red blood cells; pink). This parasite causes African sleeping sickness, a potentially fatal disease of the central nervous system. (D) Powdery mildew spores germinating on an *Arabidopsis* leaf. (E) This clump of *Ascaris* nematodes was passed rectally by a child in Africa. (A, courtesy of Robert Grant, Stephan Crainic, and James M. Hogle; B, photograph kindly provided by John Mekalanos; C, CDC, Department of Health and Human Services; D, courtesy of Kim Findlay and John Innes Centre; E, CDC/James Gathany.)



**Figure 23-3 Bacterial shapes and cell-surface structures.** (A) Bacteria are traditionally classified by shape. For example, the bacterium *Borrelia burgdorferi*, shown in Figure 23-1B, is spiral shaped and is thus classified as a spirochete. (B and C) Bacteria are also classified as *Gram positive* or *Gram negative* on the basis of a staining procedure first described in 1884. (B) Gram-positive bacteria such as *Streptococcus* and *Staphylococcus* have a single membrane and a thick cell wall made of cross-linked *peptidoglycan*. They are called Gram positive because they retain the violet dye used in the Gram-staining procedure. (C) Gram-negative bacteria such as *Escherichia coli* (*E. coli*) and *Salmonella enterica* have two membranes, separated by the *periplasm* (see Figure 11-17). The peptidoglycan cell wall of these organisms is located in the periplasm and is thinner than in Gram-positive bacteria; they therefore fail to retain the dye in the Gram-staining procedure. The inner membrane of both Gram-positive and Gram-negative bacteria is a phospholipid bilayer. The inner leaflet of the outer membrane of Gram-negative bacteria is also made primarily of phospholipids, whereas the outer leaflet of the outer membrane is composed of the unique glycosylated lipid *LPS*. (D) Cell-surface appendages are important for bacterial behavior. Both Gram-positive and Gram-negative bacteria swim using the rotation of helical flagella. An example is *V. cholerae*, depicted in panel A and shown in Figure 23-2B. Although the bacterium illustrated has only a single flagellum at one pole, many have multiple flagella. *Pili* (also called *fimbriae*) are used to adhere to various surfaces in the host and to facilitate genetic exchange between bacteria. Some kinds of pili can retract to generate force and thereby help bacteria move along surfaces.

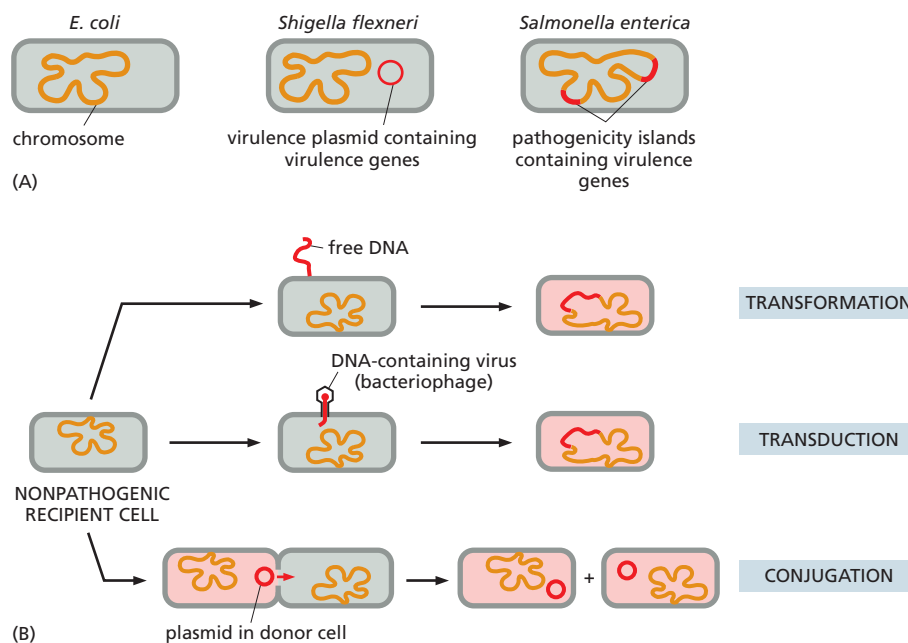
such feature is their shape—rods, spheres (*cocci*), or spirals (Figure 23-3A). They are also traditionally classified by their so-called **Gram-staining** properties, which is an older staining method that is still informative. This staining reflects differences in the structure of the bacterial surface, a layer that is important for pathogens because it directly contacts host cells and the immune system. **Gram-positive** bacteria have a thick layer of peptidoglycan cell wall outside their inner (plasma) membrane (Figure 23-3B), whereas **Gram-negative** bacteria have a thinner peptidoglycan cell wall. In both cases, the cell wall protects against lysis by osmotic swelling, and it is a target of host antibacterial proteins such as lysozyme and antibiotics such as penicillin. Gram-negative bacteria are also covered outside the cell wall by an outer membrane containing *lipopolysaccharide* (*LPS*) (Figure 23-3C). Both peptidoglycan and LPS are unique to bacteria and are recognized as *pathogen-associated molecular patterns* (*PAMPs*) by the host innate immune system, as discussed in Chapter 24. The surface of bacterial cells can also display an array of appendages, including flagella and pili, which enable bacteria to swim or adhere to desirable surfaces, respectively (Figure 23-3D). Apart from cell shape and structure, differences in ribosomal RNA and genomic DNA sequence are also used for phylogenetic classification. Because bacterial genomes are small—typically between 1,000,000 and 5,000,000 nucleotide pairs (compared to more than 3,000,000,000 for humans)—they are now simple to sequence, making this an important new classification tool.

Bacteria also exhibit extraordinary molecular, metabolic, and ecological diversity. At the molecular level, bacteria are far more diverse than eukaryotes, and they can occupy ecological niches having extremes of temperature, salt concentrations, and nutrient limitation. To describe how flexible bacterial pathogens are with respect to their growth niche, we discriminate between **facultative pathogens** and **obligate pathogens**. Facultative bacteria can replicate in an environmental reservoir such as water or soil and only cause disease if they happen to encounter a susceptible host. Obligate pathogens can only replicate inside the body of their host. Bacteria also differ in the range of hosts they will infect. A champion generalist is the opportunistic pathogen *Pseudomonas aeruginosa*, which can cause disease in a wide variety of plants and animals. In contrast, *Salmonella enterica* serovar Typhi only infects humans, causing typhoid fever, an ancient and life-threatening human disease that was sensationalized by the story of the cook “Typhoid Mary,” who carried this bacterium asymptotically and spread typhoid fever in New York City in the early 1900s.

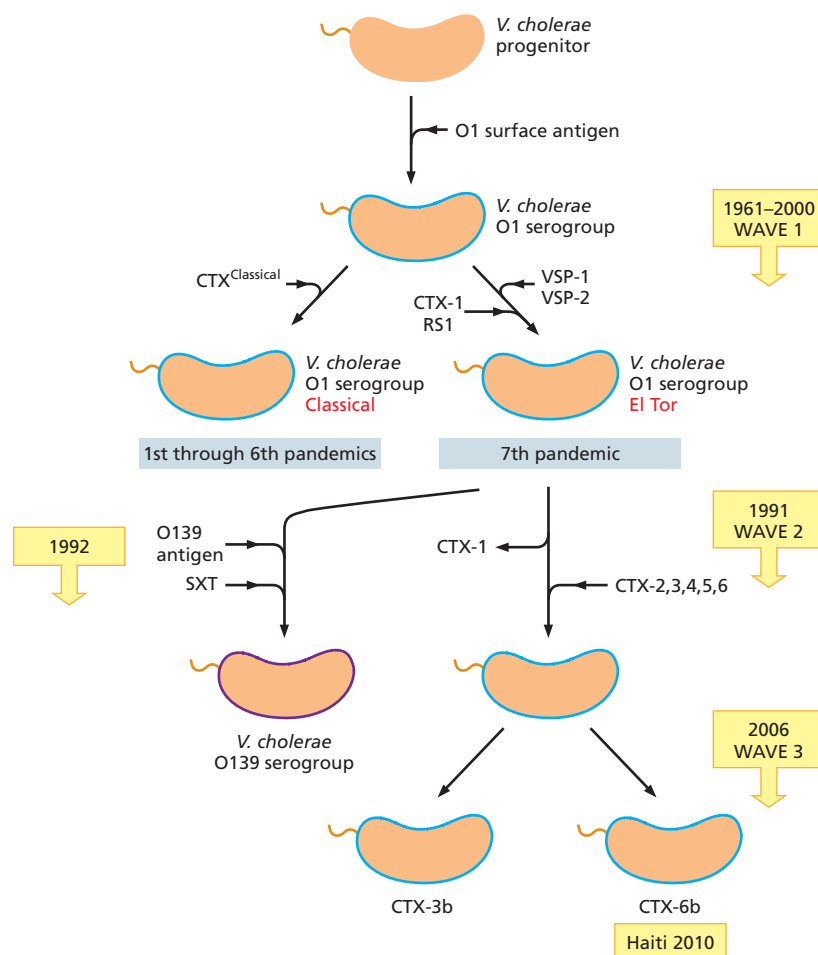
### Bacterial Pathogens Carry Specialized Virulence Genes

Genes that contribute to the ability of an organism to cause disease (as opposed to genes needed solely to grow and replicate) are called **virulence genes**, and the proteins they encode are called **virulence factors**. In general, the presence of a relatively small number of such virulence genes distinguishes pathogenic bacteria from their closest nonpathogenic relatives. Virulence genes are often clustered together on the bacterial chromosome; large clusters are called *pathogenicity islands*. Virulence genes can also be carried on *bacteriophages* (bacterial viruses) or *transposons* (see Table 5–4), both of which integrate into the bacterial chromosome, or on extrachromosomal *virulence plasmids* (Figure 23–4A).

Pathogenic bacteria are thought to emerge when groups of virulence genes are transferred together into a previously avirulent bacterium by a process called **horizontal gene transfer** (to distinguish it from vertical gene transfer from parent to offspring). Horizontal transfer can occur by one of three mechanisms: natural *transformation* by released naked DNA, *transduction* (infection) by bacteriophages, or sexual exchange by *conjugation* (Figure 23–4B and Movie 23.1). Sequencing the genomes of pathogenic and nonpathogenic bacteria has revealed that horizontal gene transfer has made important contributions to bacterial evolution, enabling species to inhabit new ecological and nutritional niches and to cause disease. Even within a single bacterial species, the amount of chromosomal



**Figure 23–4 Genetic differences between pathogenic and nonpathogenic bacteria.** (A) Genetic differences between nonpathogenic *E. coli* and two closely related food-borne pathogens—*Shigella flexneri*, which causes dysentery, and *Salmonella enterica* serovar Typhimurium, a common cause of food poisoning. Nonpathogenic *E. coli* has a single circular chromosome. The chromosome of *S. flexneri* differs from that of *E. coli* in a limited number of locations; most of the genes required for pathogenesis (virulence genes) are carried on an extrachromosomal virulence plasmid. The chromosome of *S. enterica* serovar Typhimurium carries two large inserts (pathogenicity islands) not found in the *E. coli* chromosome; these inserts each contain many virulence genes. (B) Bacterial pathogens evolve by horizontal gene transfer. This can occur by three mechanisms: natural *transformation*, in which naked DNA is taken in by competent bacteria; *transduction*, in which bacterial viruses (bacteriophages) transfer DNA from one bacterium (in the example shown, a pathogen) into another; and *conjugation* (see Movie 23.1), during which plasmid DNA, and even chromosomal DNA, is transferred from a donor to a recipient bacterium.



**Figure 23-5 Model for the evolution of pathogenic *V. cholerae* strains.**

Progenitor strains in the wild first acquired the biosynthetic pathway necessary to make the O1 antigen type of carbohydrate chain (blue outline) on the outer-membrane LPS (see Figure 23-3C), a feature that is associated with the ability to cause epidemic cholera disease. Incorporation of the CTX bacteriophage created the classical pathogenic strains responsible for the first six worldwide epidemics of cholera between 1817 and 1923. Sometime in the twentieth century, an O1 strain in the environment picked up the CTX bacteriophage again, along with an associated bacteriophage RS1 and two pathogenicity islands (VSP-1 and VSP-2), creating the El Tor (wave 1) strain that emerged as the seventh worldwide pandemic in 1961. In the 1990s, an El Tor strain was isolated that had picked up a new DNA cassette, enabling it to produce the O139 antigen type of carbohydrate chain (purple outline) rather than the O1 type. This altered the bacterium's interaction with the human immune system without diminishing its virulence; this bacterium also picked up a new pathogenicity island (SXT). Separately, wave 2 strains emerged, in which CTX-1 was replaced with CTX-2-6 bacteriophages. In 2006, wave 3 strains emerged with variant CTX-3 and CTX-6 bacteriophages. The history of the *V. cholerae* strains shown in this diagram was deduced by comparing the DNA sequences of the genome of each strain to one another and noting the key differences. An electron micrograph of *V. cholerae* is shown in Figure 23-2B.

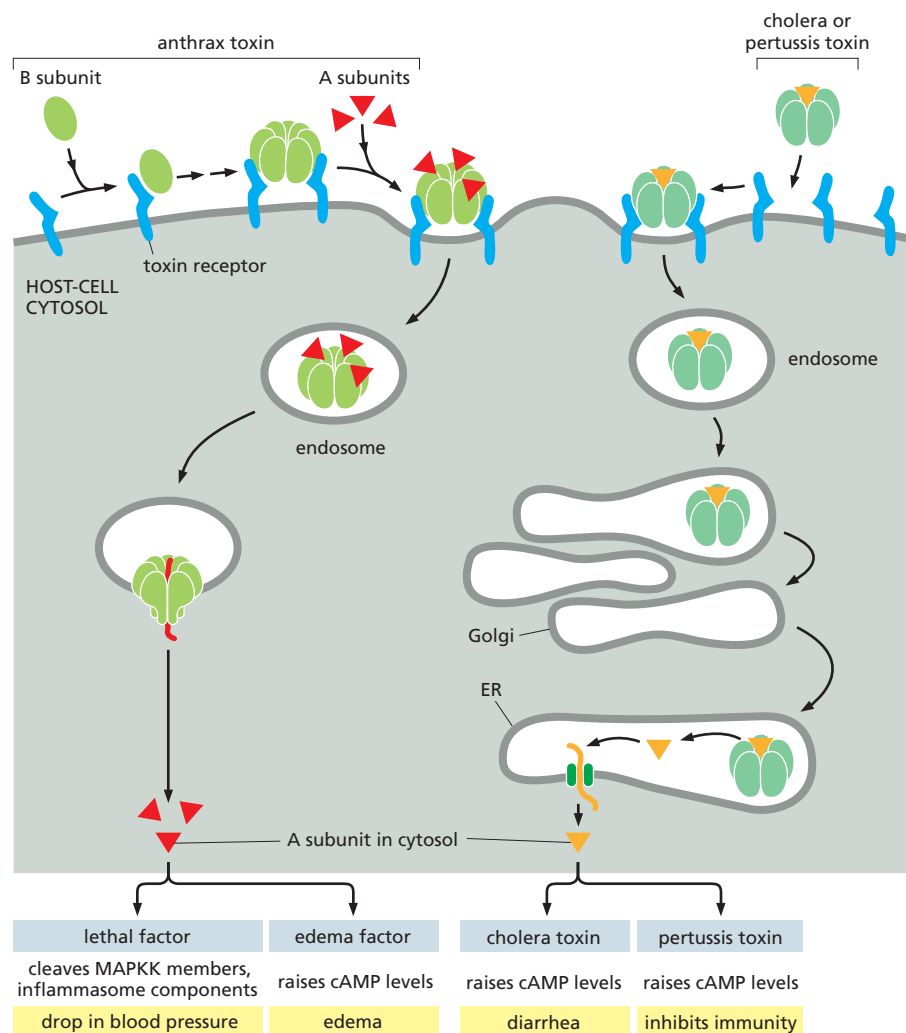
variation is astonishing; the genomes of different strains of *Escherichia coli* can differ by as much as 25%. Such variation has led to the concept that a bacterial species has both a *core genome* common to all isolates within the species and a larger *pangenome* consisting of all genes present in the full spectrum of isolates.

Acquisition of genes and gene clusters can drive the rapid evolution of pathogens and turn nonpathogens into pathogens. Consider, for example, *V. cholerae*—the Gram-negative bacterium that causes the epidemic diarrheal disease cholera. Of the hundreds of strains of *V. cholerae*, the only ones that cause pandemic human disease are those infected (see Figure 23-4B) with a mobile bacteriophage (CTX $\phi$ ; hereafter referred to as CTX) containing genes encoding the two subunits of the toxin that causes the diarrhea. As summarized in Figure 23-5, seven pandemics of *V. cholerae* have arisen since 1817. The first six were caused by the periodic reemergence of so-called *classical* strains. In addition to the toxin-encoding bacteriophage, these strains shared a similar O1 surface antigen, part of the LPS in the outer membrane (see Figure 23-3C). In 1961, the seventh pandemic began, caused by a new strain named *El Tor*, now also called a *wave 1* strain, which arose when an O1-expressing strain acquired CTX-1 and RS1 $\phi$  (hereafter referred to as RS1) bacteriophages and at least two new pathogenicity islands. El Tor strains have now displaced the classical strains. In 1991, new *wave 2* and *early wave 3* strains emerged, in which CTX-1 was replaced with CTX-2-6 bacteriophages. In 1992, another new strain emerged in which O1 was replaced with another variant called O139, which differs in a surface antigen, and thus was not recognized by antibodies present in the blood of survivors of previous cholera epidemics. Then, in 2006, current wave 3 strains emerged, which include the strain containing CTX-6b that caused the severe cholera outbreak in Haiti in 2010, infecting more than 700,000 people and leading to approximately 8500 deaths.

As this example makes clear, the rapid evolution of bacterial pathogens can be likened to an arms race that pits the survival of a bacterium against our immune systems and the tools of modern medicine. Similar struggles for survival take place between all pathogens and humans, and understanding these conflicts provides key insights into the evolution of pathogens and greatly informs how we treat new outbreaks of infectious diseases.

### Bacterial Virulence Genes Encode Toxins and Secretion Systems That Deliver Effector Proteins to Host Cells

What are the gene products that enable a bacterium to cause disease in a healthy host? The precise nature of these products differs between classes of pathogens that are largely distinguished by whether they live outside of host cells, so-called *extracellular bacterial pathogens* (discussed in this section), or inside host cells, so-called *intracellular bacterial pathogens* (discussed in a later section). For extracellular bacterial pathogens, virulence genes often encode secreted toxic proteins (*toxins*) that are released by bacteria. These toxins diffuse to target host cells, where they bind to specific receptors on the cell surface and enter the cell, interfering with host-cell proteins to elicit a response that is beneficial to the pathogen. Several of these bacterial toxins are among the most potent of known human poisons. Bacterial toxins are often composed of two protein components—an A subunit with enzymatic activity, and a B subunit that binds to host-cell receptors and directs the trafficking of the A subunit to the cytosol by various routes (Figure 23-6). The

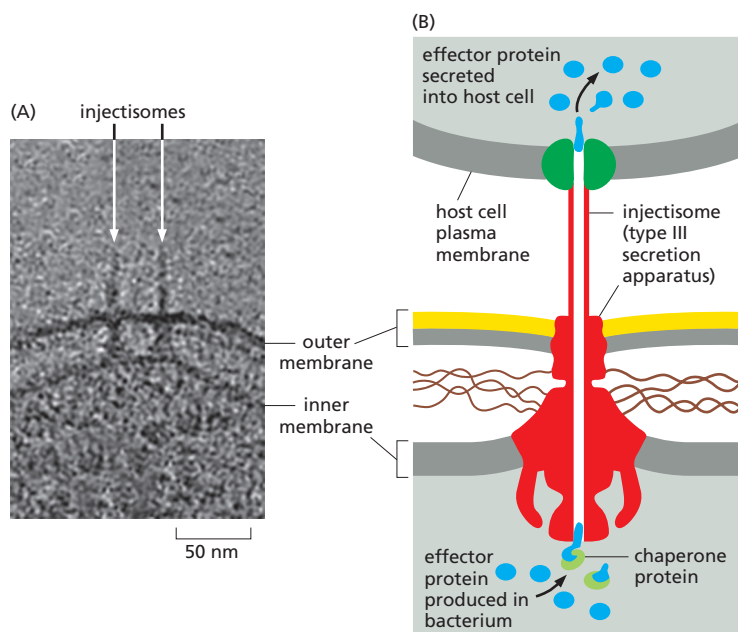


**Figure 23-6 Toxins released by bacteria bind to and enter into host cells, disrupting host-cell processes.** Bacterial toxins are often composed of A and B protein subunits. The B (binding) subunit of the toxin, which interacts with host-cell toxin receptors, can assemble after binding to toxin receptors (as with anthrax toxin) or prior to receptor binding (as with cholera and pertussis toxins). Binding to toxin receptors enables endocytosis and intracellular trafficking of the B subunits and the associated and enzymatically active A subunit(s). In the case of *Bacillus anthracis*, the B subunit changes conformation in the low-pH environment of the endosome to form a pore through which two different A subunits, lethal factor and edema factor, are transported across the membrane of the endosome in an unfolded conformation. In the cases of *V. cholerae* toxin and *Bordetella pertussis* toxin, the B and A subunits are transported to the Golgi apparatus and then to the endoplasmic reticulum (ER), where the A subunits are then translocated into the cytosol in an unfolded conformation through a protein-translocation channel (see Movie 23.2).

two subunits of **cholera toxin**, for example, are encoded by the *V. cholerae* CTX phage (see Figure 23–5), which is incorporated into the bacterial genome (**Movie 23.2**). The A subunit catalyzes the transfer of an ADP-ribose moiety from  $\text{NAD}^+$  to the trimeric G protein  $G_s$ . This ADP-ribosylation alters the G-protein  $\alpha$  subunit so that it can no longer hydrolyze its bound GTP, causing it to remain in an active state that stimulates adenylyl cyclase indefinitely (see Chapter 15). The resulting prolonged elevation in cAMP concentration within intestinal epithelial cells causes a large efflux of  $\text{Cl}^-$  and water into the gut, thereby causing the severe diarrhea that characterizes cholera. Released bacteria then contaminate food and water, spreading infection to new hosts.

Some pathogenic bacteria secrete multiple toxins, each of which targets a different signaling pathway in host cells. Anthrax, for example, is an acute infectious disease of sheep, cattle, and occasionally humans. It is caused by contact with spores of the Gram-positive bacterium *Bacillus anthracis*. Dormant spores can survive in soil for long periods. If inhaled, ingested, or rubbed into breaks in the skin, spores can germinate and the bacteria replicate. The bacteria secrete two toxins with identical B subunits but different A subunits. The B subunits bind to a host-cell surface receptor protein to transfer the two different A subunits into host cells (see Figure 23–6). The A subunits are called **lethal factor** and **edema factor**. The latter is a calmodulin-dependent adenylyl cyclase that catalyzes the production of cyclic AMP (see Figure 15–26). Injection of edema toxin into the bloodstream of an animal leads to an ion imbalance that can cause an accumulation of extracellular fluid (*edema*) in the intestine and liver. Lethal toxin is a protease that cleaves several activated members of the mitogen-activated protein kinase kinase (MAP kinase kinase; MAPKK) family (see Figure 15–50) and components of the inflammasome (see Chapter 24), disrupting intracellular signaling and leading to immune-cell dysfunction. In animals, lethal toxin targets cells of the cardiovascular system, causing shock (a large fall in blood pressure) and death.

Whereas toxins are released and diffuse to target cells, pathogens that physically contact host cells often possess specialized **contact-dependent secretion systems** that directly inject so-called *effector proteins* from the bacterium into host cells. There are several such secretion systems, and we will limit our discussion to two that are present in Gram-negative bacteria and play important roles in bacterial pathogenesis: the *type III secretion system* (Figure 23–7) and the *type IV secretion system*. The injection of effector proteins, as depicted to occur via the **type III secretion system** in Figure 23–7, can elicit a variety of



**Figure 23–7** Contact-dependent type III secretion systems can deliver effector proteins from the cytosol of a bacterium directly into the host cell. (A) Electron micrograph of type III secretion systems on the surface of a bacterial cell, each of which consists of more than two dozen proteins. (B) The large lower ring is embedded in the bacterial inner membrane, and the smaller upper ring is embedded in the bacterial outer membrane. During infection, docking of the tip of the hollow needle at a host-cell plasma membrane results in the secretion of bacterial translocator proteins (dark green), which form a pore in the host membrane. Bacterial effector proteins (blue) are initially bound to chaperone proteins in the bacterial cytosol (light green), which keep them in an unfolded conformation. Unfolded effector proteins then pass through the injectisome and are secreted into the host cell, where they fold into their active conformation. (A, Hu et al., *Cell* 168:1065–1074, 2017. With permission from Elsevier.)

host-cell responses. For extracellular pathogens, it can enable the bacterium to block phagocytosis by immune cells. For intracellular pathogens, it can promote phagocytosis by nonimmune cells, or survival inside cells. The type III and type IV secretion systems appear to have evolved independently. There is a remarkable degree of structural similarity between type III secretion systems and the bacterial flagellum (see Figures 23–2B and 23–3D), suggesting they have a common evolutionary origin. Similarly, type IV secretion systems are closely related to the conjugation apparatus that many bacteria use to exchange genetic material (see Figure 23–4B). Although the mechanism of effector protein translocation through the type III and type IV secretion systems is not well understood, the inner diameter of the type III secretion system needle is only approximately 2 nm, so proteins must be moved through in an unfolded conformation.

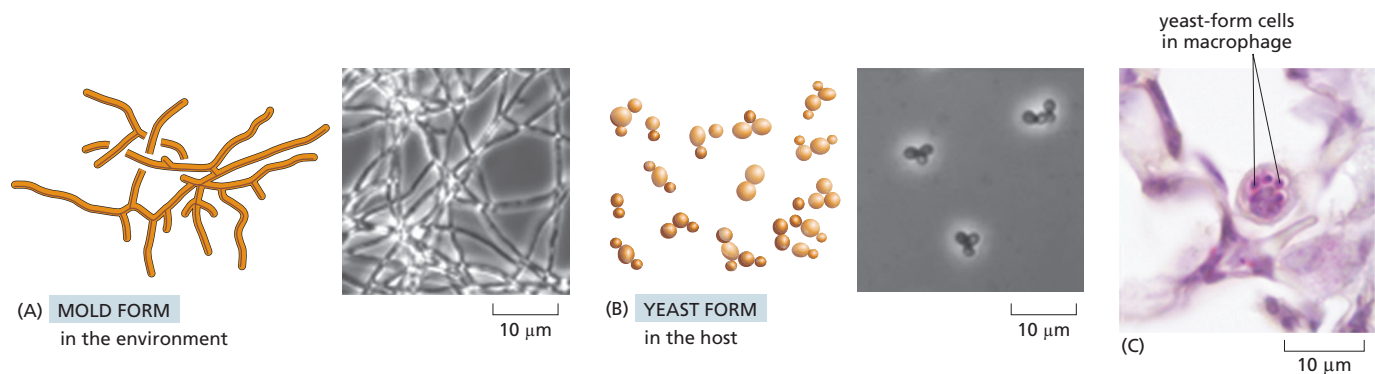
### Fungal and Protozoan Parasites Have Complex Life Cycles Involving Multiple Forms

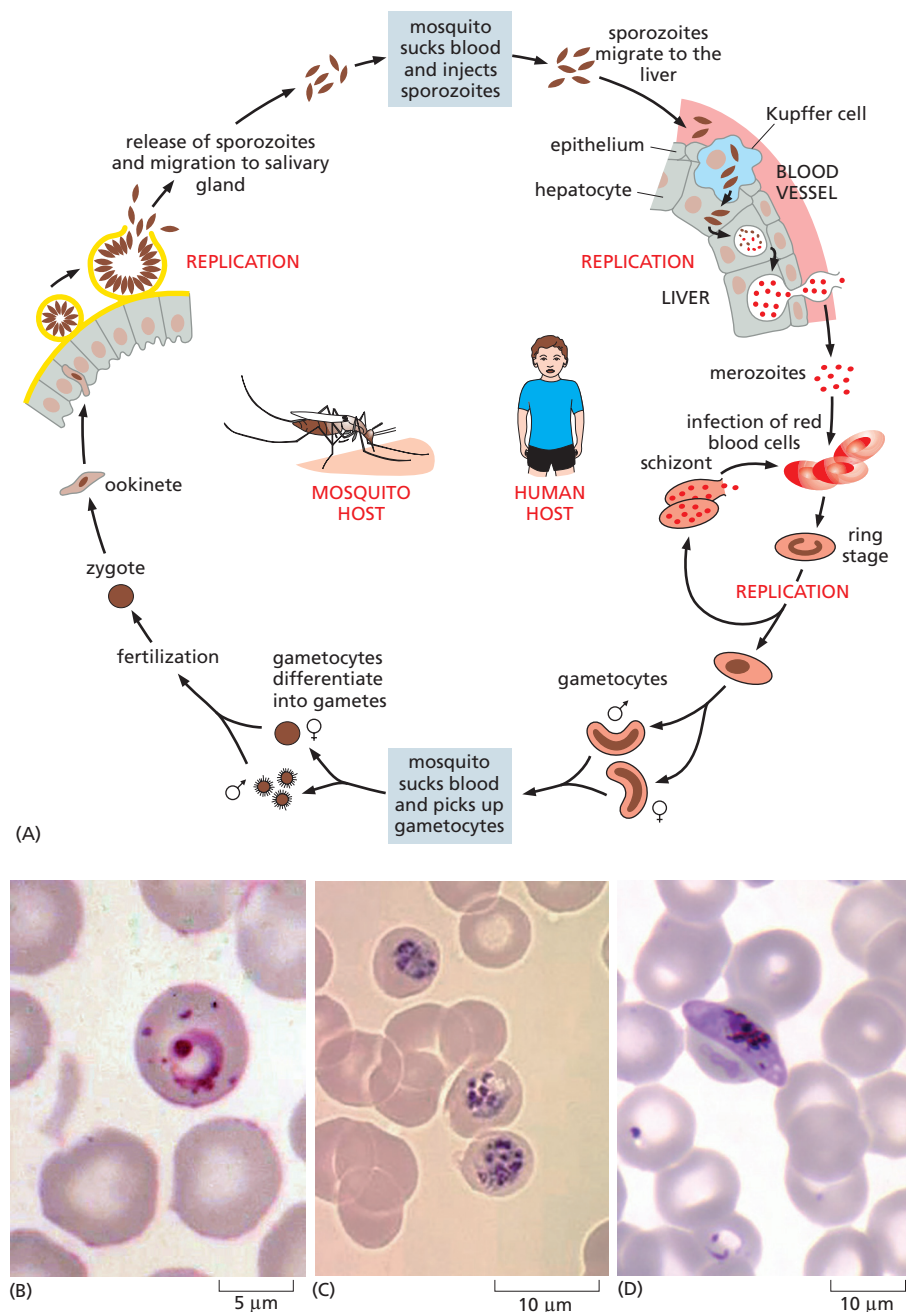
Pathogenic fungi and protozoan parasites are eukaryotes, as are their hosts. Because antifungal and antiparasitic drugs often target core molecules that are similar between parasite and host, they can be less effective and more toxic to the host than are antibiotics that target bacteria. A second characteristic of fungal and parasitic infections that makes them difficult to treat is the tendency of the pathogens to switch among several different forms during their life cycles. A drug that is effective at killing one form can be ineffective at killing another form; therefore, the population of the pathogen can survive the treatment.

**Fungi** include both unicellular *yeasts* (such as *Saccharomyces cerevisiae* and *Schizosaccharomyces pombe*, which are used to bake bread and brew beer, and as model organisms for cell biology research) and filamentous, multicellular *molds* (like those found on moldy fruit or bread; see Figure 23–2D). Most of the important pathogenic fungi exhibit *dimorphism*—the ability to grow in either yeast or mold form. The yeast-to-mold or mold-to-yeast transition is frequently associated with infection. *Histoplasma capsulatum*, for example, grows as a mold at low temperature in the soil, but it switches to a yeast form when inhaled into the lung, where it can cause the disease histoplasmosis (Figure 23–8). The most common human fungal pathogen, *Candida albicans*, which often exists as a resident of the human oral and gastrointestinal microbiota and is also an opportunistic pathogen that can cause serious disease in immunocompromised people by entering the bloodstream and rapidly dividing.

**Protozoan parasites** are single-celled eukaryotes with more elaborate life cycles than fungi, and they frequently require more than one host. **Malaria** is the most devastating protozoal disease, infecting more than 200 million people every year and killing upward of 400,000. It is caused by four species of *Plasmodium*, which are transmitted to humans by the bite of the female *Anopheles* mosquito. *Plasmodium falciparum* causes the most serious form of malaria and is the most intensively studied of the malaria-causing parasites. It exists in many distinct

**Figure 23–8 Dimorphism in the pathogenic fungus *Histoplasma capsulatum*.** (A) At low temperature in the soil, *H. capsulatum* grows as a multicellular filamentous mold consisting of many individual cells connected together. (B) After it is inhaled into the lung of a mammal, the increase in temperature to 37°C causes a switch to a yeast form consisting of small clumps of round cells. (C) A stained histologic section of a mouse lung infected with *H. capsulatum*, showing a macrophage containing yeast forms of the pathogen. (A and B, courtesy of Sinem Beyhan and Anita Sil; C, courtesy of Davina Hocking Murray and Anita Sil.)





**Figure 23-9** The complex life cycle of malaria parasites. (A) The sexual cycle of *Plasmodium falciparum* requires passage between a human host and an insect host (see Movie 23.3). (B–D) Blood smears from people with malaria, showing three different forms of the parasite that appear in red blood cells: B, ring stage; C, schizont; and D, gametocyte. (B–D, courtesy of the Centers for Disease Control, Division of Parasitic Diseases, DPDx.)

forms, and it requires both the human and mosquito hosts to complete its sexual cycle (Figure 23-9A and Movie 23.3). Several of these forms are highly specialized to invade and replicate in specific tissues—the lining of the insect gut, the human liver, and the human red blood cell. Even within a single host-cell type, the red blood cell, the *Plasmodium* parasite undergoes a complex sequence of developmental events, reflected in striking morphological changes (Figure 23-9B, C, and D).

### All Aspects of Viral Propagation Depend on Host-Cell Machinery

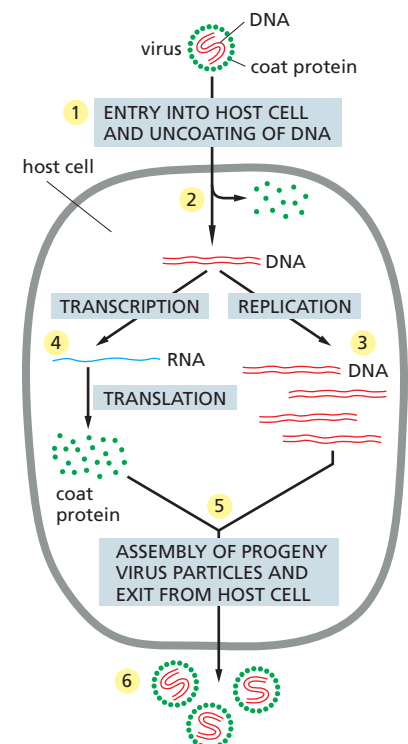
Bacteria, fungal, and protozoan pathogens are themselves living cells. They use their own machinery for DNA replication, transcription, and translation, and, for the most part, they provide their own sources of metabolic energy from nutrients in their environment. **Viruses**, by contrast, are the ultimate hitchhikers, carrying little more than genetic information in the form of nucleic acid. Many clinically

TABLE 23–1 Viruses That Cause Human Disease

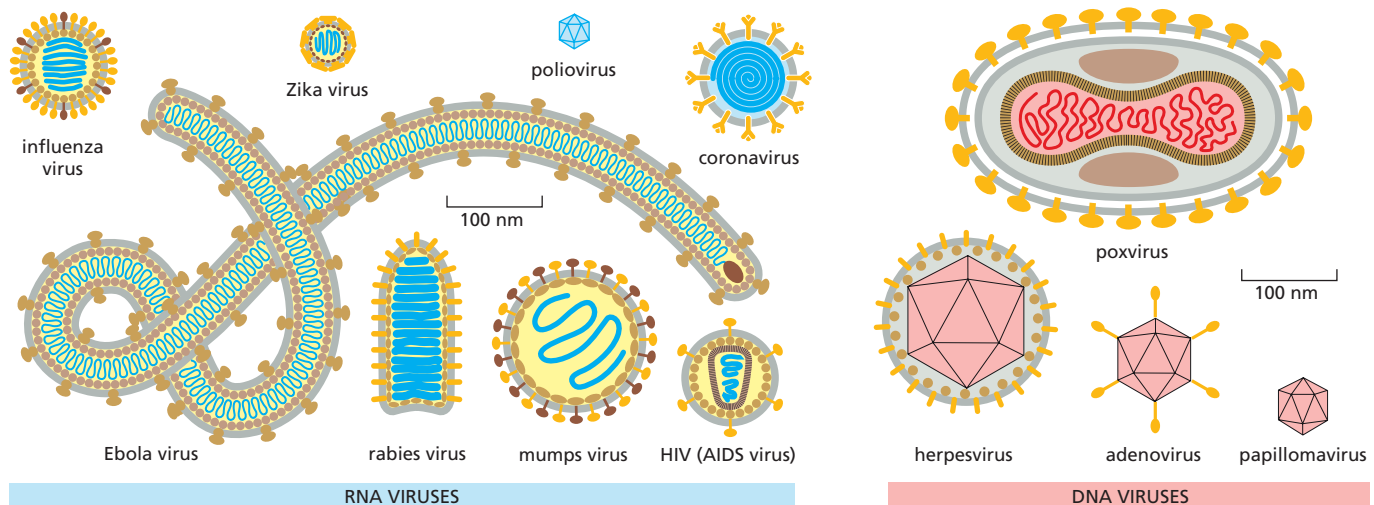
| Virus                              | Genome type                            | Disease                                    |
|------------------------------------|----------------------------------------|--------------------------------------------|
| Adenovirus                         | Double-stranded DNA                    | Respiratory disease                        |
| Epstein–Barr virus (EBV)           | Double-stranded DNA                    | Infectious mononucleosis                   |
| Herpes simplex virus 1             | Double-stranded DNA                    | Recurrent cold sores                       |
| Human papillomavirus               | Double-stranded DNA                    | Warts, cancer                              |
| Varicella-zoster virus             | Double-stranded DNA                    | Chickenpox and shingles                    |
| Smallpox virus (Variola)           | Double-stranded DNA                    | Smallpox                                   |
| Hepatitis-B virus                  | Part single-, part double-stranded DNA | Hepatitis B                                |
| Human immunodeficiency virus (HIV) | Single-stranded RNA [+] strand         | Acquired immune deficiency syndrome (AIDS) |
| Coronavirus                        | Single-stranded RNA [+] strand         | Common cold, respiratory disease, COVID-19 |
| Hepatitis-A virus                  | Single-stranded RNA [+] strand         | Hepatitis A                                |
| Hepatitis-C virus                  | Single-stranded RNA [+] strand         | Hepatitis C                                |
| Poliovirus                         | Single-stranded RNA [+] strand         | Poliomyelitis                              |
| Rhinovirus                         | Single-stranded RNA [+] strand         | Common cold                                |
| Yellow fever virus                 | Single-stranded RNA [+] strand         | Yellow fever                               |
| Zika virus                         | Single-stranded RNA [+] strand         | Zika virus disease                         |
| Ebola virus                        | Single-stranded RNA [–] strand         | Ebola virus disease                        |
| Influenza virus type A             | Single-stranded RNA [–] strand         | Respiratory disease (flu)                  |
| Measles virus                      | Single-stranded RNA [–] strand         | Measles                                    |
| Mumps virus                        | Single-stranded RNA [–] strand         | Mumps                                      |
| Rabies virus                       | Single-stranded RNA [–] strand         | Rabies                                     |

important human viruses have relatively small genomes consisting of double-stranded DNA or single-stranded RNA [either positive (+) sense RNA that can be directly translated into proteins or negative (–) sense RNA that must be replicated to produce (+) sense RNA] (Table 23–1). We now have complete genome sequences of almost all of them.

Viral genomes typically encode three types of protein: proteins for replicating the genome, proteins for packaging the genome and delivering it to more host cells, and proteins for modifying the structure or function of the host cell to enhance the replication of the virus. In general, viral replication involves the following sequence of steps: (1) entry into the host cell, (2) disassembly of the infectious virus particle, (3) replication of the viral genome, (4) transcription of viral genes and synthesis of viral proteins, (5) assembly of these viral components into progeny virus particles, and (6) release of progeny virions (Figure 23–10). A single virus particle (a *virion*) that infects a single host cell can produce thousands of progeny.



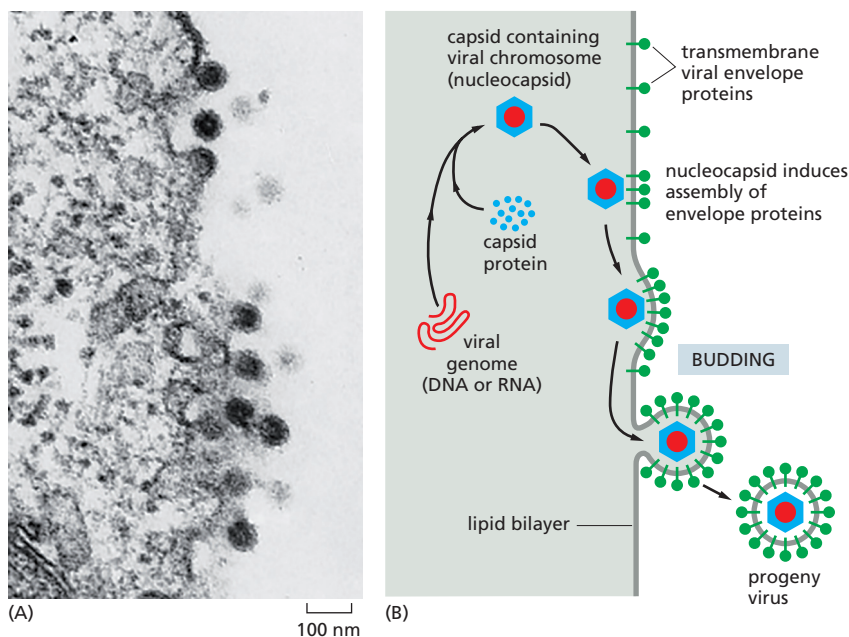
**Figure 23–10 A simple viral life cycle.** The hypothetical simple virus shown here consists of a small double-stranded DNA molecule that codes for only a single viral capsid protein. To reproduce, the viral genome must (1) enter a host cell, where it (2) disassembles and releases its genetic material. The genome is (3) replicated to produce multiple copies, and (4) transcribed and translated to produce the viral coat protein. The viral genomes can then (5) assemble spontaneously with the coat protein to form a new virus particle, which (6) escapes from the host cell. No known virus is this simple.



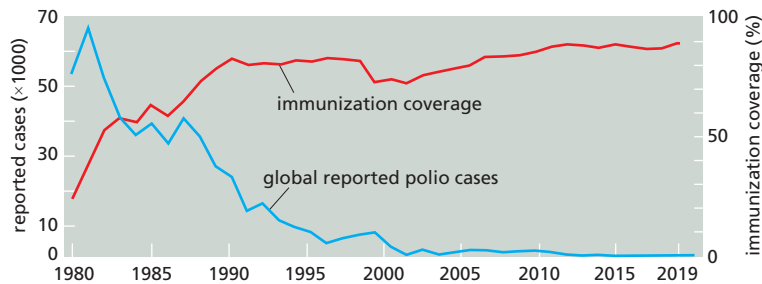
Virions come in a wide variety of shapes and sizes (Figure 23–11), and although most have relatively small genomes, genome size can vary considerably. A giant virus of amoebae, called *pithovirus*, was recently revived from a 30,000-year-old ice core harvested from permafrost in Siberia and is the largest known virus by physical size, with 1.5- $\mu\text{m}$ -long particles and a double-stranded DNA genome of 610,000 nucleotides. The virions of *poxviruses* are also large: they are 250–350 nm long and enclose a genome of double-stranded DNA of about 270,000 nucleotides. At the other end of the size scale are the virions of *parvovirus*, which are less than 30 nm in diameter and have a single-stranded DNA genome of fewer than 5000 nucleotides.

Viral genomes are packaged in a protein coat, called a **capsid**, which in some viruses is further enclosed by a lipid bilayer membrane, or envelope. The capsid is made of one or several proteins, arranged in regular arrays that often produce structures with either helical symmetry, which results in a cylindrical structure (for example, influenza, measles, and bunyavirus), or icosahedral symmetry (for example, poliovirus and herpesvirus; see Figure 23–11). Other viruses instead produce capsids with more complicated or irregular structures (for example, poxviruses and Ebola virus; see Figure 23–11). A capsid packaged with the viral

**Figure 23–11 Examples of viral morphology.** As shown, both DNA and RNA viruses vary greatly in both size and shape. Although there is a general correlation between virus physical size and genome size, there are likely to be outliers such as Ebola virus, which has a large physical size but a smaller genome size suggesting a lower degree of genome compaction.



**Figure 23–12 Acquisition of a viral envelope.** (A) Electron micrograph of an animal cell from which several copies of an enveloped virus (*Semliki Forest virus*) are budding. (B) Schematic drawing of the envelope assembly and budding processes. The lipid bilayer that surrounds the viral capsid is derived directly from the plasma membrane of the host cell. In contrast, the proteins in this lipid bilayer (shown in green) are encoded by the viral genome. (A, from A. Loewy et al., *J. Virol.* 69:469–475, 1995. Reproduced with permission from the American Society for Microbiology.)



**Figure 23-13 Effective control of a viral disease through vaccination.** The graph shows the number of cases of poliomyelitis reported per year worldwide (blue line) and the global immunization coverage as a percentage of the human population (red line). As immunization coverage increased, disease incidence decreased. (Figure based on information found at <https://polioeradication.org>.)

genome is called a *nucleocapsid*. The nucleocapsids of *nonenveloped viruses* usually leave an infected cell by lysing it. For *enveloped viruses*, by contrast, the nucleocapsid is enclosed within a lipid bilayer membrane that the virus acquires in the process of budding from the host-cell plasma membrane, which it does without disrupting the membrane or killing the cell (Figure 23-12).

Because the host cell performs most of the critical steps in viral replication, the identification of effective antiviral drugs that do not harm the host can be difficult. The most effective strategy for containing viral diseases is through vaccination of potential hosts. For example, a highly successful vaccination program eradicated smallpox infection from the planet by 1980. Vaccines effective against *poliovirus*, which causes poliomyelitis and paralysis, have existed since the 1950s, and a more recent global vaccination initiative, begun in 1988, seeks to completely eradicate poliovirus infection. The number of poliomyelitis cases worldwide has declined from an estimated 350,000 in 1988, to 176 in 2019, and there is hope of eradication in the coming years (Figure 23-13).

## Summary

*Infectious diseases are caused by pathogens, which include viruses, bacteria, and fungi, as well as protozoan and metazoan parasites. All pathogens must have mechanisms for entering their host and for evading immediate destruction by the host. Pathogenic bacteria produce specific virulence factors that mediate their interactions with the host; these proteins change the behavior of host cells in ways that promote the replication and spread of the bacteria. Eukaryotic pathogens such as fungi and protozoan parasites typically pass through several different forms during the course of infection; the ability to switch among these forms is usually required for these pathogens to survive in a host and cause disease. In some cases, such as malaria, parasites must pass sequentially through several host species to complete their life cycles. Unlike bacteria and eukaryotic parasites, viruses have no metabolism of their own and no intrinsic ability to produce the proteins encoded by their DNA or RNA genomes; they rely on subverting the machinery of their host cell.*

## CELL BIOLOGY OF PATHOGEN INFECTION

The mechanisms through which pathogens cause disease are extremely diverse. Nonetheless, all pathogens must carry out certain common tasks: they must gain access to the host, reach an appropriate growth niche, avoid host defenses, replicate, and exit from the infected host to spread to an uninfected one. In this section, we examine the cell-biological strategies that many pathogens use to accomplish these tasks. We describe how pathogens initially gain access to the host by overcoming epithelial barriers and colonizing epithelia. We then recount how extracellular pathogens disturb host cells without entering them to establish a niche and avoid host defenses. We next cover the myriad ways in which intracellular pathogens, the master manipulators of host-cell biology, enter host cells, reach an appropriate niche in the cytosol or in a membrane-bound compartment, avoid or alter membrane traffic, mobilize the cytoskeleton, manipulate autophagy, and take over metabolism. Finally, we cover how pathogen evolution shapes the immune avoidance and drug resistance.

## Pathogens Breach Epithelial Barriers to Infect the Host

The first step in infection is for the pathogen to gain access to the host. A thick covering of skin protects most parts of the human body from the environment. The protective boundaries of some other human tissues (eyes, nasal passages, respiratory tract, mouth, digestive tract, urinary tract, and female genital tract) are less robust. In the lungs and small intestine, for example, the barrier is just a single monolayer of epithelial cells. Nonetheless, all these epithelia serve as barriers to infection.

Wounds in barrier epithelia are one way for pathogens to gain direct access to unoccupied niches within otherwise sterile host tissues. This avenue of entry requires little in the way of pathogen specialization, and some pathogens can cause serious illness if they enter through such wounds. Staphylococci from the skin and nose and streptococci from the throat and mouth are two examples of opportunistic bacterial pathogens that can be residents of the normal microbiota (described in more detail in the final section of this chapter) in certain individuals, yet are also responsible for many serious infections resulting from breaches in epithelial barriers. The emergence of bacterial strains of *Staphylococcus* that are resistant to the antibiotics commonly used for treatment (for example, methicillin-resistant *Staphylococcus aureus*, or MRSA, which can cause serious skin and tissue infections) is of particular concern. Papillomaviruses, which cause warts and cervical cancer, also take advantage of breaches in epithelial barriers.

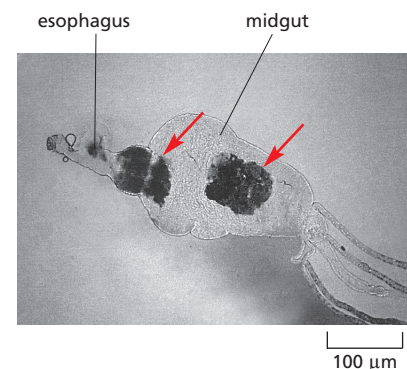
Another efficient way for a pathogen to cross the skin is to catch a ride in the saliva of a biting arthropod. A diverse group of bacteria, viruses, and protozoa has developed the ability to survive in arthropods that they use as *vectors* for transmission to a mammalian host. Many *zoonoses*, a term that refers to diseases spread to humans from other animals, are spread in this way. As discussed earlier, the *Plasmodium* protozoan that causes malaria develops through several forms in its life cycle, including some that are specialized for survival in a human and others that are specialized for survival in a mosquito vector (see Figure 23–9). Viruses that are spread by mosquito bites also cause yellow fever, dengue fever, as well as Zika virus disease—the latter garnered worldwide attention during an epidemic in 2015–2016, and although Zika infection is often asymptomatic in adults, in pregnant women it can cause birth defects in the fetus. These three viruses replicate in both insect cells and mammalian cells, as required for their transmission by an insect vector.

The efficient spread of a pathogen via an insect vector requires that an individual insect consumes a blood meal from an infected host and transfers the pathogen to a nonimmune host. In a few striking cases, the behavior of the insect is altered by the pathogen so that its transmission to a new host is more likely. An example is the bacterium *Yersinia pestis*, which causes bubonic plague. It multiplies in the flea's foregut to form aggregated masses that physically block the digestive tract; during each repeated, but futile, attempt at feeding, some of the bacteria in the foregut are flushed into the bite site, thus transmitting plague to a new host (Figure 23–14).

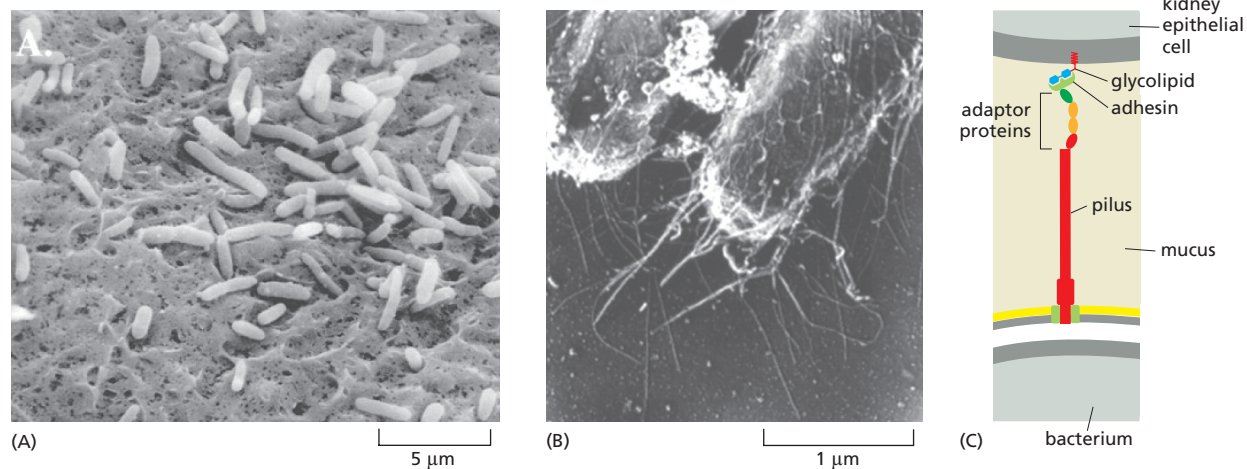
## Pathogens That Colonize an Epithelium Must Overcome Its Protective Mechanisms

Epithelial barriers such as the skin lining of the mouth and large intestine are densely populated by the microbiota, whereas others, including the lining of the lower lung and the bladder, are more sparsely populated. Nevertheless, these epithelial barriers have mechanisms to prevent excessive microbial colonization. A layer of protective mucus covers the respiratory epithelium, and the coordinated beating of motile cilia sweeps the mucus and trapped bacteria up and out of the lung. The epithelial lining of the bladder and the upper gastrointestinal tract also has a thick layer of mucus, and these organs are periodically flushed by urination and by peristalsis, respectively, which washes away microbes.

Pathogens that infect these epithelial surfaces, including pathogenic bacteria and eukaryotic parasites, have evolved specific features to overcome these protective mechanisms. Those that infect the urinary tract, for example, adhere



**Figure 23–14 Plague bacteria within a flea.** This light micrograph shows the digestive tract dissected from a flea that had dined about 2 weeks previously on the blood of an animal infected with the plague bacterium, *Yersinia pestis*. The bacteria multiplied in the flea gut to produce large cohesive aggregates (red arrows); the bacterial mass on the left is occluding the passage between the esophagus and the midgut. This type of blockage prevents a flea from digesting its blood meals, so that hunger causes it to bite repeatedly, disseminating the infection. (From B.J. Hinnebusch, E.R. Fischer, and T.G. Schwan, *J. Infect. Dis.* 178:1406–1415, 1998.)

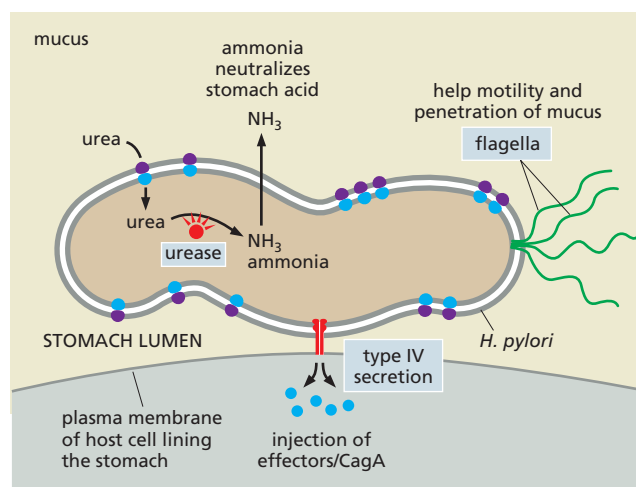


tightly to the epithelial lining via specific **adhesins**, which are proteins or protein complexes that recognize and bind to cell-surface molecules on the epithelium. An important group of adhesins in *E. coli* strains that infect the kidney are components of *pili*—surface projections that can be several micrometers long and thus able to span the thickness of the protective mucus layer; at the tip of each pilus is an adhesin protein that binds tightly to the D-galactose–D-galactose disaccharide on glycolipids on the surface of kidney cells (Figure 23-15). Strains of *E. coli* that infect the bladder rather than the kidney express a second kind of pilus with a different adhesin protein that binds to D-mannose–decorated proteins on bladder epithelial cells. It is the specificity of the adhesin proteins on the tips of the two types of pili that is responsible for the bacteria colonizing different parts of the urinary tract.

The epithelial lining of the stomach is an especially hostile environment for microbes. Besides the thick layer of mucus and peristaltic washing, the acidic pH (average pH  $\approx$  2) is lethal to almost all bacteria ingested in food. Yet, the stomach is home to resident microbial species including *H. pylori*, which persists for life as part of the stomach microbiota of approximately half of all humans on Earth. Although it does not cause disease in most individuals, *H. pylori* can cause stomach ulcers and cancers. The hypothesis that a persistent bacterial infection could cause stomach ulcers was initially met with skepticism. The young Australian doctor who made the initial discovery finally proved the point: he drank a pure culture of bacteria and developed inflammation of the stomach, which often precedes the development of ulcers. Antibiotics can now effectively cure a patient of recurrent ulcers. *H. pylori* has several adaptations that allow it to colonize the harsh environment in the stomach (Figure 23-16). One is to use its flagella

**Figure 23-15 Pathogenic *E. coli* in the infected bladder of a mouse.**

(A) Scanning electron micrograph of uropathogenic *E. coli*, a common cause of bladder and kidney infections. The bacteria are attached to the surface of epithelial cells lining the infected bladder. (B) A close-up view of one of the bacteria showing the pili on its surface. (C) An *E. coli* pilus spans the mucus layer and has adaptor proteins on its tip that bind to glycolipids on the surface of kidney cells. (A, from G.E. Soto and S.J. Hultgren, *J. Bacteriol.* 181:1059–1071, 1999. With permission from the American Society for Microbiology; B, from D.G. Thanassi and S.J. Hultgren, *Methods* 20:111–126, 2000. With permission from Elsevier.)



**Figure 23-16 *H. pylori* interaction with epithelial cells in the stomach lining.**

These bacteria colonize the gastric epithelium by using their flagella to swim through the mucus layer lining the stomach and attach to the surface of epithelial cells. *H. pylori* produces the enzyme urease, which converts urea obtained from the environment into ammonia that is released and neutralizes acid surrounding the bacterium, raising the pH in its local environment. *H. pylori* also uses its type IV secretion system to secrete the effector protein CagA, which assists in colonization but also causes chronic disruption of host-cell pathways and prolonged inflammation, which predisposes infected individuals to gastric ulcers and cancer.

for chemotactic motility, allowing it to penetrate mucus and seek out the more neutral pH near the surface of gastric epithelial cells. Another is to produce the enzyme *urease*, which converts urea to ammonia to neutralize the acid in its immediate vicinity.

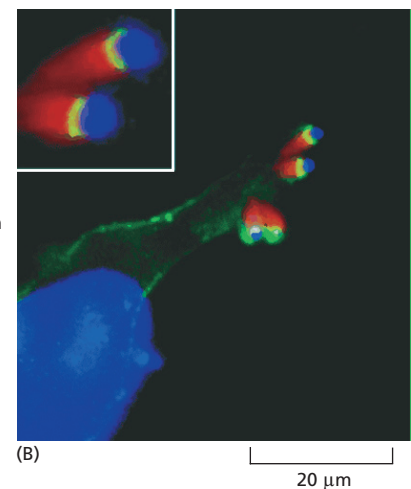
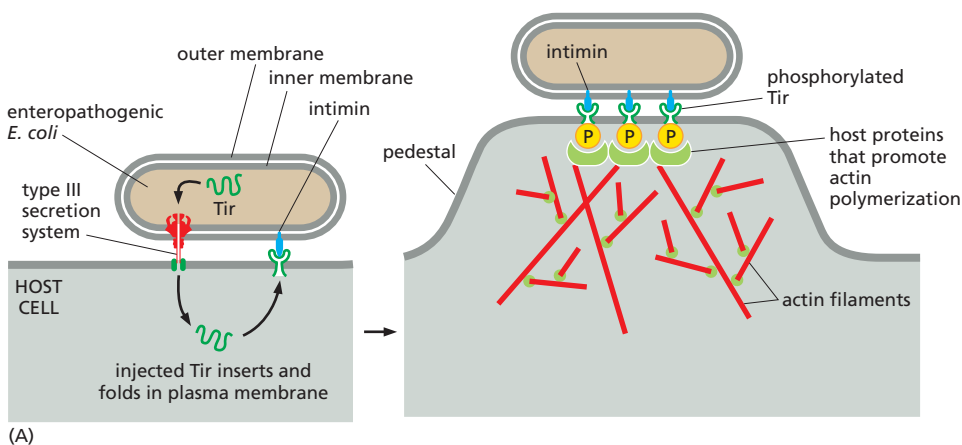
### Extracellular Pathogens Use Toxins and Contact-dependent Secretion Systems to Disturb Host Cells Without Entering Them

**Extracellular pathogens** adhere to epithelial and other surfaces without entering their underlying host cells. Many such pathogens exert effects on the host by secreting toxins that diffuse to target cells. For example, *Bordetella pertussis*, the bacterium that causes whooping cough, colonizes the respiratory epithelium and circumvents the normal mechanism that clears the respiratory tract by expressing adhesins that bind ciliated epithelial cells. The adherent bacteria also produce toxins that eventually kill the ciliated cells, compromising the host's ability to clear the infection. The most familiar of these is *pertussis toxin*, which, like the cholera toxin discussed earlier (both are shown in Figure 23-6), has an A subunit that ADP-ribosylates the  $\alpha$  subunit of the G protein  $G_i$ , causing the G protein to remain in the inactive GDP-bound state and preventing it from inhibiting the activity of the host cell's adenylyl cyclase, thereby increasing the production of cyclic AMP. This toxin also interferes with the chemotactic pathway that neutrophils use to seek out and destroy bacteria (see Figures 16-3 and 16-81). *B. pertussis* colonization of the respiratory tract causes severe coughing, which helps spread the infection.

Other extracellular pathogens use contact-dependent secretion systems to directly inject bacterial effector proteins into the host cells to which they adhere. An example is *H. pylori*, which infects the stomach epithelium as described above. The *H. pylori* genome contains a pathogenicity island (see Figure 23-4A) that encodes a type IV secretion system and the effector protein CagA, which is injected by the secretion system into the host cell (Figure 23-16). CagA helps the bacterium persist in the stomach by affecting cell movement, inducing inflammation, altering host gene expression, changing cell proliferation and apoptosis, and disrupting cell-cell junctions. However, chronic alteration of these pathways is also a predisposing factor for gastric ulcers and cancer.

Another example is enteropathogenic *E. coli* (EPEC), which causes diarrhea and can be lethal to young children. EPEC uses a type III secretion system (see Figure 23-7) to deliver its own special receptor protein (called *Tir*) into the plasma membrane of a host intestinal epithelial cell (Figure 23-17 and Movie 23.4). The extracellular domain of Tir binds to the bacterial surface protein *intimin*, triggering actin polymerization in the host cell that results in the formation of a unique cell-surface protrusion called a *pedestal*; this pushes the tightly adherent bacteria up about 1–5  $\mu\text{m}$  from the host-cell membrane, thereby promoting bacterial

**Figure 23-17 Interaction of enteropathogenic *E. coli* (EPEC) with host intestinal epithelial cells.** (A) When EPEC contacts an epithelial cell in the lining of the human gut, it delivers a bacterial protein called Tir into the host cell through a type III secretion system. Tir then inserts into the plasma membrane of the host cell, where it functions as a receptor for the bacterial adhesin protein intimin. Next, a host-cell protein tyrosine kinase phosphorylates the intracellular domain of Tir on tyrosines. Phosphorylated Tir recruits host-cell proteins [including an adaptor protein, a nucleation-promoting factor (NPF), and the Arp2/3 complex] that trigger actin polymerization (see Figure 16-12). Consequently, a branched network of actin filaments assembles underneath the bacterium, forming an actin pedestal (see Movie 23.4). (B) EPEC on a pedestal. In this fluorescence micrograph, the DNA of the EPEC and host cell is labeled in blue, Tir protein is labeled in green, and host-cell actin filaments are labeled in red. The inset shows a close-up view of the two upper bacteria on pedestals. (B, from D. Goosney et al., *Annu. Rev. Cell Dev. Biol.* 16:173–189, 2000. With permission from Annual Reviews.)



movement along the cell surface by a mechanism described later for the actin-based motility of intracellular pathogens. A similar strategy is used by vaccinia virus (the virus that was used as a vaccine to eradicate smallpox) to form mobile actin-rich membrane extensions, which promote spread of the virus from cell to cell. The study of how EPEC and vaccinia virus promote actin polymerization has been of major importance in understanding how intracellular signaling pathways regulate the cytoskeleton in normal, uninfected cells (discussed in Chapter 16). Although actin mobilization promotes the spread of these pathogens, the symptoms of EPEC infection (severe diarrhea) are caused by the loss of absorptive microvilli and disruption of signaling pathways in epithelial cells, which are triggered by Tir and other secreted bacterial effector proteins.

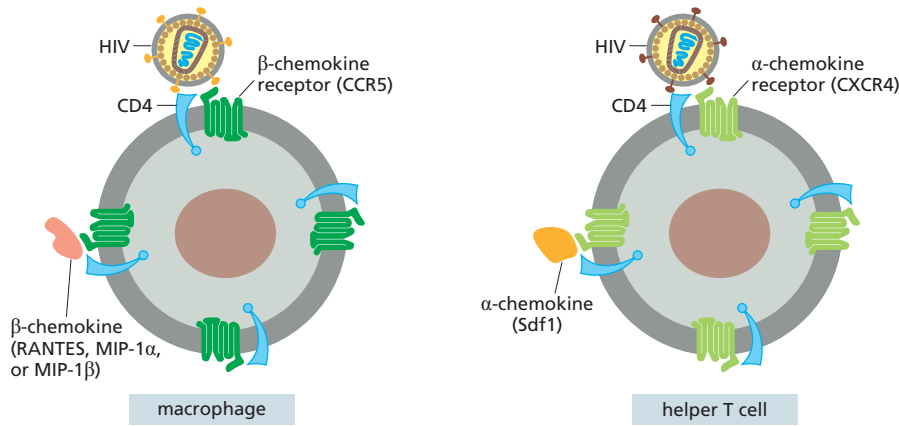
### Intracellular Pathogens Have Mechanisms for Both Entering and Leaving Host Cells

**Intracellular pathogens** have to cross barriers, adhere to host cells, and also enter host cells to cause disease. These include all viruses and many bacteria and protozoa. Each of these has a preferred niche for replication and survival within host cells. Bacteria and protozoa replicate either in the cytosol or within a membrane-enclosed compartment. While most RNA viruses replicate within the cytosol, most DNA viruses replicate in the nucleus (poxviruses are a notable exception). Life inside a host cell has several advantages. The pathogens are not accessible to *antibodies*, nor are they easy targets for phagocytic cells (discussed in Chapter 24); furthermore, intracellular bacteria and protozoa are bathed in a rich source of nutrients, and viruses have access to the host cell's biosynthetic machinery for their reproduction. This lifestyle, however, requires that the pathogen have mechanisms for entering host cells, for finding a suitable subcellular niche where it can replicate, and for exiting from the infected cell to spread the infection. Below we consider some of the myriad ways that individual intracellular pathogens exploit and modify host-cell biology to satisfy these requirements.

### Viruses Bind to Virus Receptors at the Host-Cell Surface

The first step in infection for any intracellular pathogen is to bind to the surface of the host target cell. Viruses accomplish this by the binding of viral surface proteins to **virus receptors**, which are cell-surface proteins that perform various functions in uninfected cells and have been co-opted as binding sites for viral proteins. The first virus receptor identified was an *E. coli* surface protein that is recognized by the bacteriophage lambda; the protein normally functions to transport the sugar maltose from outside the bacterium to the inside where it is used as an energy source. Receptors need not be proteins, however: an envelope protein of herpes simplex virus, for example, binds to heparan sulfate proteoglycans (discussed in Chapter 19) on the surface of certain vertebrate host cells, and simian virus 40 (SV40) binds to a glycolipid. The specificity of virus-receptor interactions often serves as a barrier preventing the spread of a virus from one species to another. Acquiring the ability to bind to a new receptor often requires multiple changes in a virus, but it can be crucial in allowing the cross-species transmission that can result in new disease outbreaks.

Viruses that infect animal cells generally exploit cell-surface receptor molecules that are either ubiquitous [such as angiotensin-converting enzyme 2 (ACE2) used by the coronavirus SARS-CoV-2] or are found uniquely on those cell types in which the virus replicates (such as the neuron-specific proteins used by rabies virus). Although a virus usually uses a single type of host-cell receptor, some viruses use more than one type. An important example is HIV, which requires two types of receptors to enter a host cell. Its primary receptor is CD4, a cell-surface protein on helper T cells and macrophages that is involved in immune recognition (discussed in Chapter 24). It also requires a co-receptor, which is either CCR5 (a receptor for  $\beta$ -chemokines) or CXCR4 (a receptor for  $\alpha$ -chemokines), depending on the particular variant of the virus; macrophages are susceptible



**Figure 23-18 Receptor and co-receptors for HIV.** All strains of HIV require the CD4 protein as a primary receptor. Early in an infection, most of the viruses use CCR5 as a co-receptor, allowing them to infect macrophages and their precursors, monocytes. As the infection progresses, mutant variants of the virus arise that now use CXCR4 as a co-receptor, enabling them to infect helper T cells efficiently. The natural ligand for the chemokine receptors (Sdf1 for CXCR4; RANTES, MIP-1 $\alpha$ , or MIP-1 $\beta$  for CCR5) blocks co-receptor function and prevents viral invasion.

only to HIV variants that use CCR5 for entry, whereas helper T cells are most efficiently infected by variants that use CXCR4 (**Figure 23-18**). The viruses that are found within the first few months after HIV infection almost invariably use CCR5, which explains why individuals that carry an altered *CCR5* gene are less susceptible to HIV infection. In the later stages of infection, viruses often either switch to use CXCR4 or adapt to use both co-receptors through the accumulation of viral mutations. In this way, the virus can change the cell types it infects as the disease progresses. It may seem paradoxical that viruses would infect immune cells, as we might expect that virus binding would trigger an immune response; but invasion of an immune cell can be a useful way for a virus to weaken the immune response and travel around the body to infect other immune cells.

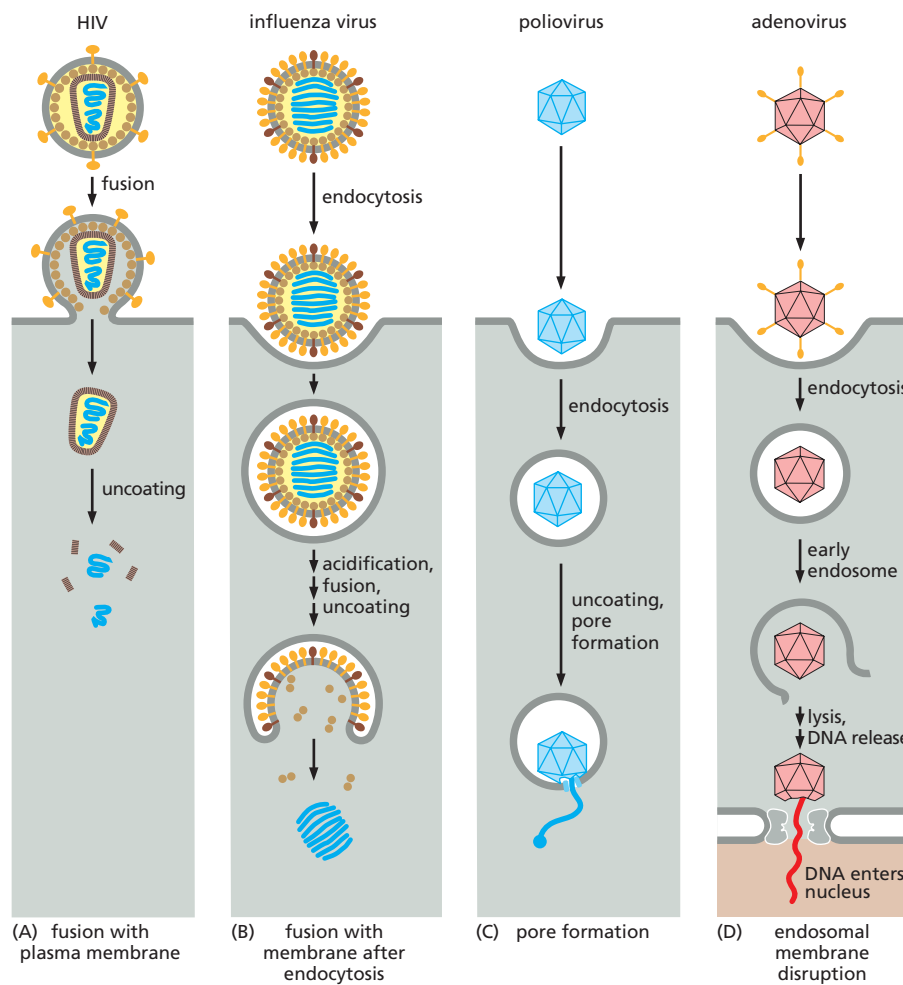
### Viruses Enter Host Cells by Membrane Fusion, Pore Formation, or Membrane Disruption

After recognition and attachment to the host-cell surface, viruses must enter into the cell for viral replication to proceed. Entry into host cells requires that viruses overcome different challenges that depend on virus size and structure (see **Figure 23-11**). **Enveloped viruses** must regulate membrane fusion processes, both to ensure that their membrane envelopes fuse only with the appropriate host-cell membrane and to prevent fusion with one another. Such regulation is achieved by the coronavirus SARS-CoV-2, for example, by requiring both binding of the virus spike protein to the ACE2 receptor and cleaving of the spike protein by a host-cell protease to enable fusion with the host-cell membrane.

Some enveloped viruses, such as HIV, enter the host cell by fusing their envelope membrane at neutral pH with the plasma membrane (**Figure 23-19A**). In this scenario, binding to receptors or co-receptors usually triggers a conformational change in a viral envelope protein that exposes a normally buried fusion peptide (discussed in Chapter 13).

Most enveloped (and nonenveloped) viruses enter cells by activating signaling pathways in the cell that induce endocytosis, commonly via clathrin-coated pits (see **Figure 13-7**), leading to internalization into endosomes. Large viruses that do not fit into clathrin-coated vesicles, such as poxviruses, often enter cells by *macropinocytosis*, a process by which membrane ruffles fold over and entrap fluid into macropinosomes (see **Figure 13-68**).

Once inside endosomes, enveloped viruses such as influenza A virus must fuse their envelope with the endosomal membrane from the luminal side, and they frequently sense the acid environment in the late endosome as a cue to trigger a conformational change in a viral surface protein that exposes a fusion peptide (**Figure 23-19B** and **Movie 23.5**). The mechanism of membrane fusion mediated by viral spike glycoproteins has similarities with SNARE-mediated membrane fusion during normal vesicular trafficking (discussed in Chapter 13). The H<sup>+</sup> pumped into the early endosome also has another effect; it enters the influenza virion through an ion channel in the viral envelope and triggers changes in the



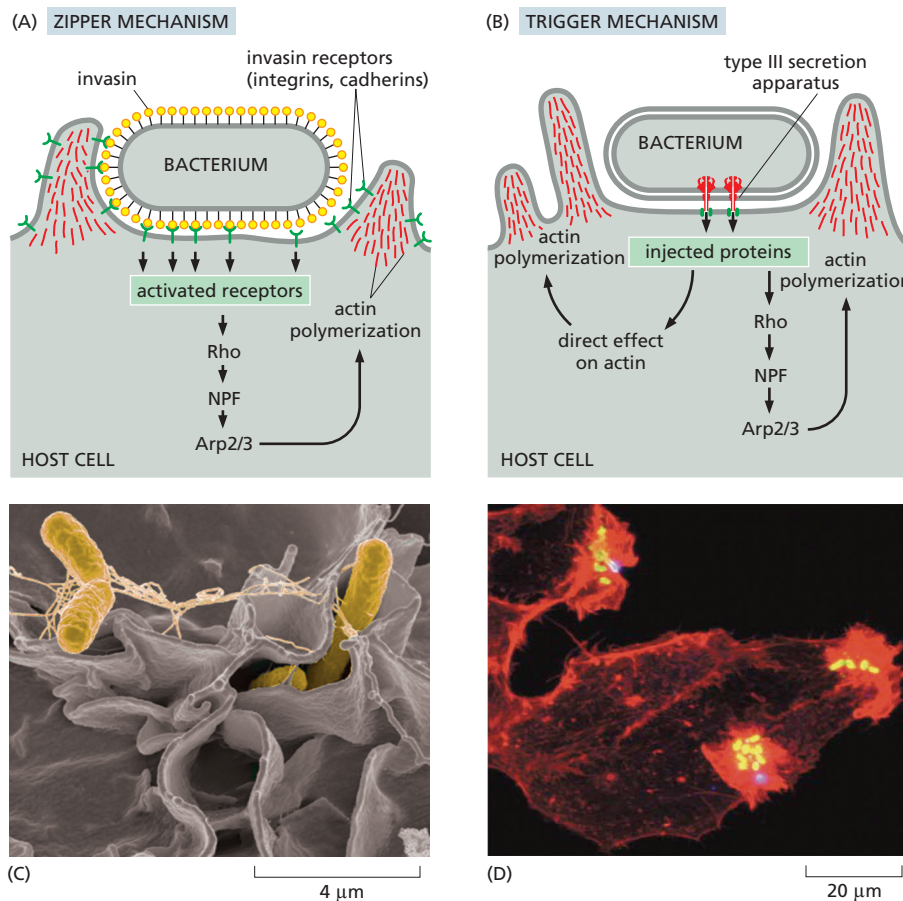
**Figure 23-19 Four virus entry strategies that follow virus binding to virus receptors.** (A) Some enveloped viruses, such as HIV, fuse directly with the host-cell plasma membrane to release their RNA genome (blue) and capsid proteins (brown) into the cytosol. (B) Other enveloped viruses, such as influenza virus, first bind to cell-surface receptors, triggering receptor-mediated endocytosis; when the endosome acidifies, the virus envelope fuses with the endosomal membrane, releasing the viral RNA genome (blue) and capsid proteins (tan) into the cytosol. (C) Poliovirus, a nonenveloped virus, induces receptor-mediated endocytosis, and then forms a pore in the endosomal membrane to extrude its RNA genome (blue) into the cytosol. (D) Adenovirus, another nonenveloped virus, uses a more complicated strategy: it induces receptor-mediated endocytosis and then disrupts the endosomal membrane, releasing the capsid including its DNA genome into the cytosol. The trimmed-down virus eventually docks onto a nuclear pore and releases its DNA (red) directly into the nucleus where it is transcribed and replicated (see Movie 23.5).

viral capsid. These priming steps allow the capsids to disassemble once released into the cytosol after virus fusion with the late endosomal membrane.

Because they lack a surrounding lipid bilayer, **nonenveloped viruses** enter host cells in a fundamentally different way. Poliovirus binds to a cell-surface receptor, triggering both receptor-mediated endocytosis (see Figure 13-54) and a conformational change in the viral particle. The conformational change exposes a hydrophobic projection on one of the capsid proteins, which inserts into the endosomal membrane to form a pore. The viral RNA genome then enters the cytosol through the pore, leaving the capsid in the endosome (Figure 23-19C). *Adenovirus* disrupts the endosomal membrane after it is taken up by receptor-mediated endocytosis, releasing the remainder of the virus into the cytosol. During endosomal trafficking and subsequent transport within the cytosol, adenoviruses undergo multiple uncoating steps, which sequentially remove structural proteins and ready the virus particles to release their DNA into the nucleus through nuclear pore complexes (Figure 23-19D).

### Bacteria Enter Host Cells by Phagocytosis

Bacteria are much larger than viruses—too large to be taken up either through pores or by receptor-mediated endocytosis. Instead, they enter host cells by phagocytosis, which is a normal function of phagocytes such as neutrophils, macrophages, and dendritic cells (discussed in Chapter 24). These phagocytes patrol the tissues of the body and ingest and destroy microbes; however, some intracellular bacterial pathogens such as *M. tuberculosis* use this to their advantage and have evolved to survive and multiply inside macrophages.

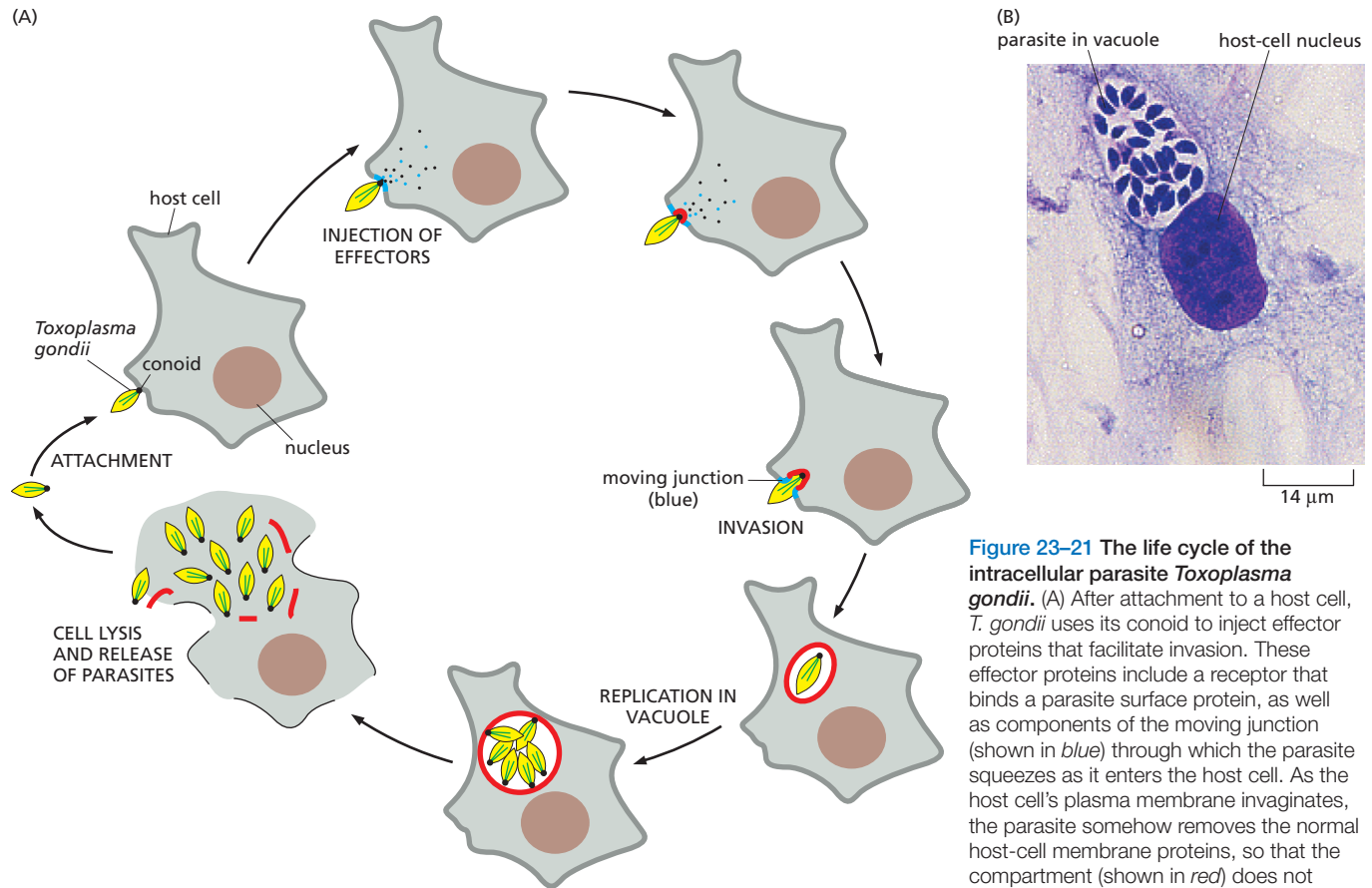


**Figure 23-20** Mechanisms used by bacteria to induce phagocytosis by host cells that are normally nonphagocytic.

(A) In the zipper mechanism, bacteria express an invasion protein that binds with high affinity to a host-cell receptor, which is often a cell-cell or cell-matrix adhesion protein. (B) In the trigger mechanism, bacteria inject a set of effector molecules into the host-cell cytosol through a type III secretion system called SPI1 (*Salmonella* pathogenicity island 1), inducing membrane ruffling. Both the zipper and trigger mechanisms cause the polymerization of actin at the site of bacterial attachment by activating Rho family small GTPases and the Arp2/3 complex. (C) A scanning electron micrograph showing *Salmonella enterica* invasion by the trigger mechanism. Bacteria (pseudocolored yellow) are shown surrounded by a small membrane ruffle. (D) Fluorescence micrograph showing that the large ruffles that engulf the *Salmonella* bacteria are actin rich. The bacteria are labeled in green and actin filaments in red; because of the color overlap, the bacteria appear yellow. (C, from Rocky Mountain Laboratories, NIAID, NIH; D, from J.E. Galán, *Annu. Rev. Cell Dev. Biol.* 17:53–86, 2001. With permission from Annual Reviews.)

Some bacterial pathogens can invade host cells that are normally nonphagocytic. One way they do so is by expressing an invasion protein that binds with high affinity to a host-cell receptor, which is often a cell-cell or cell-matrix adhesion protein (discussed in Chapter 19). For example, *Yersinia pseudotuberculosis* (a bacterium that causes diarrhea and is a close relative of the plague bacterium *Y. pestis*) expresses a protein called invasins that has an RGD motif that is similar to fibronectin's and which likewise is recognized by host-cell  $\beta_1$ -integrins (see Figure 19-48). *Listeria monocytogenes*, which causes a rare but serious form of food poisoning, invades host cells by expressing a protein that binds to the cell-cell adhesion protein E-cadherin (see Figures 19-4 and 19-5). For both these bacterial species, binding of the bacterial invasion proteins to the host-cell adhesion proteins stimulates signaling through members of the Rho family of small GTPases (discussed in Chapter 16). This in turn activates NPFs and the Arp2/3 complex, leading to actin polymerization at the site of bacterial attachment. Actin polymerization, sometimes accompanied by the assembly of a clathrin coat, drives the advancement of the host cell's plasma membrane over the adhesive surface of the microbe, resulting in the phagocytosis of the bacterium—a process known as the *zipper mechanism* of invasion (Figure 23-20A).

A second pathway by which bacteria can invade nonphagocytic cells is known as the *trigger mechanism* (Figure 23-20B). It is used by various pathogens, including the food-borne pathogen *Salmonella enterica* serovar Typhimurium, and it is initiated when the bacterium injects a set of effector molecules into the host-cell cytosol through a type III secretion system (see Figure 23-7). Some of these effector molecules activate Rho family proteins, which in turn stimulate actin polymerization, as just discussed. Other bacterial effector proteins directly interact with host-cell cytoskeletal elements, nucleating and stabilizing actin filaments and causing the rearrangement of actin cross-linking proteins. The overall effect is to cause the formation of ruffles on the surface of the host cell (Figure 23-20C and D),



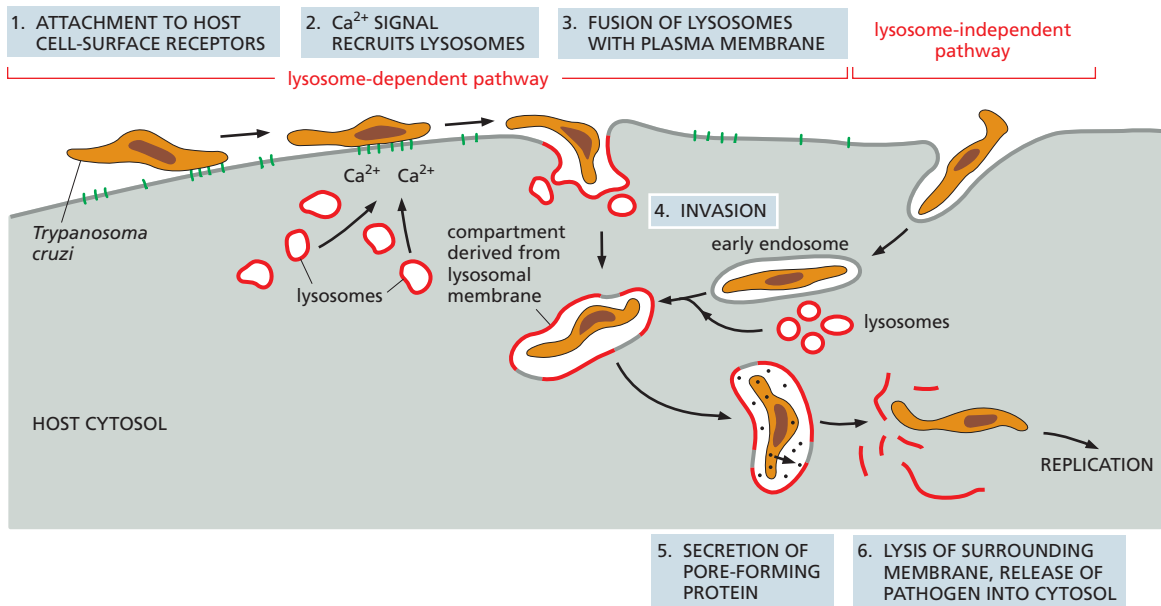
**Figure 23-21** The life cycle of the intracellular parasite *Toxoplasma gondii*. (A) After attachment to a host cell, *T. gondii* uses its conoid to inject effector proteins that facilitate invasion. These effector proteins include a receptor that binds a parasite surface protein, as well as components of the moving junction (shown in *blue*) through which the parasite squeezes as it enters the host cell. As the host cell's plasma membrane invaginates, the parasite somehow removes the normal host-cell membrane proteins, so that the compartment (shown in *red*) does not fuse with lysosomes. After several rounds of replication, the parasite causes the compartment to break down and the host cell to lyse, releasing the progeny parasites to infect other host cells (see Movie 23.6). (B) Light micrograph of *T. gondii* replicating within a membrane-enclosed compartment (a parasitophorous vacuole) in a cultured cell. (B, courtesy of Manuel Camps and John Boothroyd.)

which fold over and engulf the bacteria by a process that resembles macropinocytosis (see Figure 13-68). The appearance of cells being invaded by use of the trigger mechanism is similar to the ruffling induced by some extracellular growth factors, suggesting that the bacteria hijack normal intracellular signaling pathways.

### Intracellular Eukaryotic Parasites Actively Invade Host Cells

The uptake of viruses and bacteria into host cells is carried out largely by the host, with the pathogen being a relatively passive participant. In contrast, intracellular eukaryotic parasites, which are typically much larger than other types of intracellular pathogens, invade host cells through a variety of complex pathways that usually require energy expenditure by the parasite.

*Toxoplasma gondii*, a cat parasite that also causes occasional serious human infections in pregnant and immunocompromised individuals, is an example. *T. gondii* invasion involves the secretion of parasite effector proteins that target host-cell components, as well as the activities of the parasite's own cytoskeleton (Figure 23-21 and Movie 23.6). When this protozoan contacts a host cell, it protrudes an unusual microtubule-based structure called a *conoid*, which facilitates host-cell entry. At the point of contact, the parasite discharges effector proteins into the host cell from specialized secretory organelles. One of these effector proteins is a receptor that inserts into the host-cell plasma membrane and binds to a parasite surface protein. Other effector proteins form a ring-like *moving junction* through which the parasite squeezes using forces generated by its own unusual actin and myosin cytoskeleton. Remarkably, as the parasite invades it removes host transmembrane proteins from the surrounding membrane, so that it is eventually protected in a membrane-enclosed compartment, or *parasitophorous vacuole*, that does not fuse with lysosomes and does not participate in host-cell membrane trafficking processes. The specialized membrane is selectively porous: it allows the parasite to take up small metabolic intermediates and nutrients from



the host cell's cytosol but excludes macromolecules. Malaria parasites invade human red blood cells using a similar mechanism.

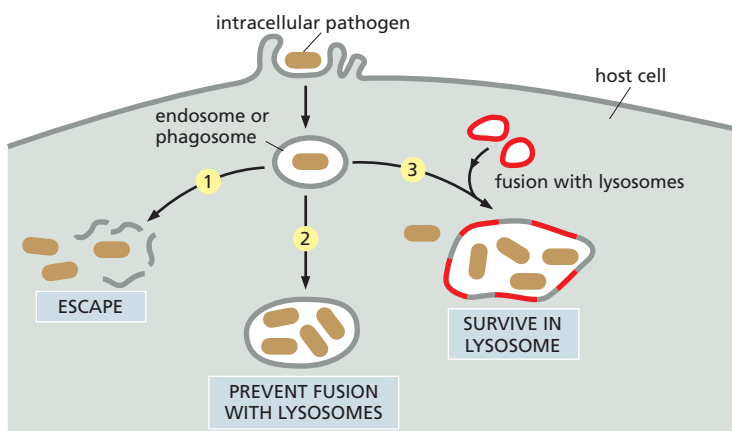
The protozoan *Trypanosoma cruzi*, which causes Chagas disease in Mexico and Central and South America, uses two alternative invasion strategies. In a *lysosome-dependent pathway*, the parasite attaches to the host's cell-surface receptors, inducing a local increase in  $\text{Ca}^{2+}$  in the host cell's cytosol. The  $\text{Ca}^{2+}$  signal recruits lysosomes to the site of parasite attachment, and the lysosomes fuse with the host cell's plasma membrane, allowing the parasites rapid access to the lysosomal compartment (Figure 23-22). In a *lysosome-independent pathway*, the parasite penetrates the host-cell plasma membrane by inducing the membrane to invaginate, without lysosome recruitment.

### Some Intracellular Pathogens Escape from the Phagosome into the Cytosol

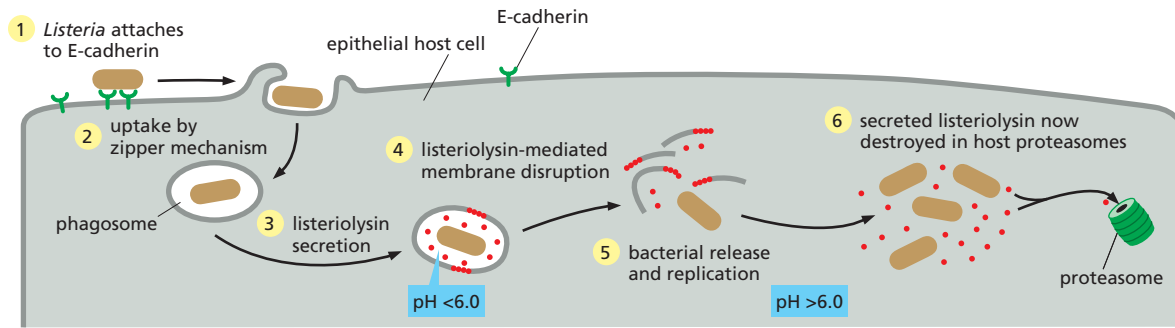
All intracellular pathogens, including viruses, bacteria, and eukaryotic parasites, face a similar problem: they must find a compartment within the host cell where they can replicate themselves. After their endocytosis by a host cell, they usually find themselves in an endosomal compartment, which normally would fuse with lysosomes (see Figure 13-67)—a dangerous place for pathogens because of the presence of many antimicrobial factors. To survive, pathogens use a variety of strategies (Figure 23-23). Some escape from the endosomal compartment before such fusion. Others remain in the endosomal compartment but modify it so that

**Figure 23-22** The two alternative strategies that *Trypanosoma cruzi* uses to invade host cells. In the lysosome-dependent pathway (left), *T. cruzi* recruits host-cell lysosomes to its site of attachment to the host cell.

The lysosomes fuse with the invaginating plasma membrane to create an intracellular compartment constructed almost entirely of lysosomal membrane. After a brief stay in the compartment, the parasite secretes a pore-forming protein that disrupts the surrounding membrane, thereby allowing the parasite to escape into the host-cell cytosol and proliferate. In the lysosome-independent pathway (right), the parasite induces the host plasma membrane to invaginate and pinch off without recruiting lysosomes; then, lysosomes fuse with the endosome prior to the parasite's escape into the cytosol.



**Figure 23-23** Choices that an intracellular pathogen faces. After entry into a host cell, generally through phagocytosis into a membrane-enclosed compartment, intracellular pathogens can use one of three strategies to survive and replicate. Pathogens that escape into the cytosol (1) include all viruses, *Trypanosoma cruzi*, *Listeria monocytogenes*, and *Shigella flexneri*. Those that prevent fusion with lysosomes (2) include *Mycobacterium tuberculosis*, *Legionella pneumophila*, and *Toxoplasma gondii*. Those that survive in the lysosome (3) include *Salmonella enterica*, *Coxiella burnetii*, and *Leishmania*.



it no longer fuses with lysosomes. Still others have evolved to weather the harsh conditions in the lysosome.

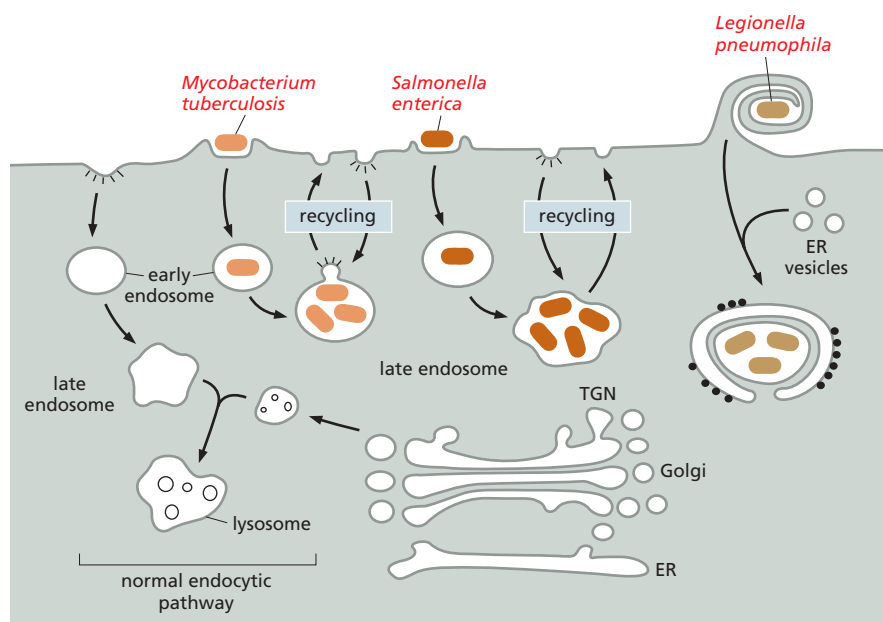
*Trypanosoma cruzi* uses the escape strategy by secreting a pore-forming toxin that lyses the lysosome membrane, releasing the parasite into the host cell's cytosol (see Figure 23-22). The bacterium *Listeria monocytogenes* uses a similar strategy. After phagocytosis by the zipper mechanism, it secretes a protein called *listeriolysin O*, which disrupts the phagosomal membrane, releasing the bacteria into the cytosol (Figure 23-24).

### Many Pathogens Alter Membrane Traffic in the Host Cell to Survive and Replicate

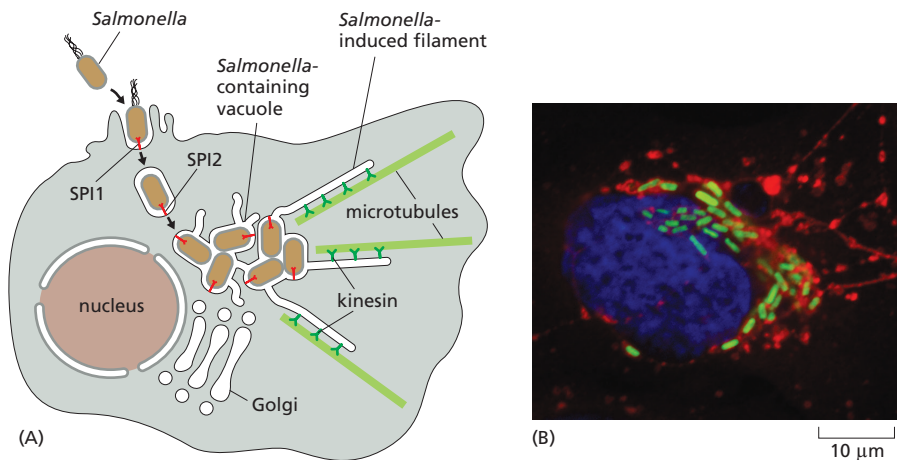
Intracellular pathogens that remain in membrane-bound compartments of the host cell after invasion (see Figure 23-23) often alter these compartments to reduce their exposure to antimicrobial molecules and allow for pathogen survival and reproduction. To do so, these pathogens must alter membrane (vesicular) traffic, for example by slowing or preventing normal fusing of endosomes with lysosomes (Figure 23-25).

Different pathogens have distinct strategies for altering host-cell membrane traffic. *M. tuberculosis* prevents the maturation of the early endosome that contains the bacteria, so the endosome never acidifies or acquires the other characteristics of a late endosome or lysosome. Phagosomes containing *Salmonella enterica*, in contrast, acidify and acquire markers of late endosomes and lysosomes, but the bacteria slow the process of phagosomal maturation. They do so by injecting effector proteins through a second type III secretion system, distinct from that

**Figure 23-24 Escape of *Listeria monocytogenes* by selective destruction of the phagosomal membrane.** The bacterium attaches to E-cadherin or other receptors on the surface of host epithelial cells and induces its own uptake by the zipper mechanism (see Figure 23-20A). Within the phagosome, the bacterium secretes the protein listeriolysin O, which is activated at pH < 6 and forms oligomers in the phagosome membrane, thereby creating large pores and eventually disrupting the membrane. Once in the host-cell cytosol, the bacteria begin to replicate and continue to secrete listeriolysin O; because the pH in the cytosol is > 6, however, the listeriolysin O there is less active and is also rapidly degraded by proteasomes. Thus, the host cell's plasma membrane remains intact.



**Figure 23-25 Modifications of membrane traffic in host cells by bacterial pathogens to slow or prevent normal fusing of endosomes with lysosomes.** Intracellular bacterial pathogens, including *Mycobacterium tuberculosis*, *Salmonella enterica*, and *Legionella pneumophila*, all replicate in membrane-enclosed compartments, but the compartments differ. *M. tuberculosis* remains in a compartment that has early endosomal markers and continues to communicate with the plasma membrane via transport vesicles. *S. enterica* replicates in a compartment that has late endosomal markers. *L. pneumophila* replicates in an unusual compartment that is wrapped in rough endoplasmic reticulum (ER) membrane and communicates with the ER via transport vesicles. TGN = trans Golgi network.



**Figure 23-26** *Salmonella enterica* residing in a modified phagosomal compartment called the *Salmonella*-containing vacuole.

These bacteria invade the host cell using one of two type III secretion systems to inject effector proteins that induce the trigger mechanism of microbe entry illustrated in Figure 23-20B. (A) After its engulfment into a phagosome, the bacterium inactivates the first type III secretion system and activates the second type III secretion system to inject different effector proteins, which remodel the phagosome into the specialized *Salmonella*-containing vacuole. One of the injected effector proteins activates host kinesin motor proteins to pull membrane tubules outward toward the plus ends of the microtubules (see Figure 16-53), forming a structure with an unusual tubulated shape that is of unknown functional significance. (B) Fluorescence micrograph showing *S. enterica* in a *Salmonella*-containing vacuole. The bacteria are stained green, the microtubules red, and the nucleus blue. (B, courtesy of Stephane Meresse.)

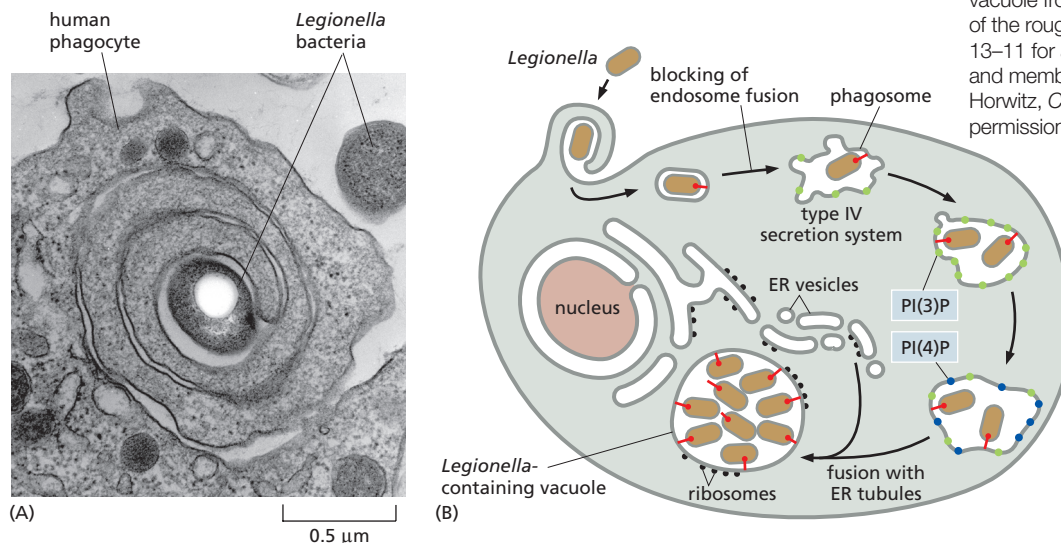
involved in invasion by the trigger mechanism (see Figure 23-20). Some of these bacterial effectors activate host kinesin motor proteins to pull membrane tubules outward from the phagosome along cytoplasmic microtubules, forming a specialized compartment called the *Salmonella*-containing vacuole (Figure 23-26).

Other bacteria seem to find shelter in intracellular compartments that are distinct from those of the usual endocytic system. One example is *Legionella pneumophila*, which was first recognized as a human pathogen in 1976, when it was found to be the cause of a type of pneumonia known as **Legionnaires' disease**. *L. pneumophila* is normally a parasite of freshwater amoebae but can be spread to humans by central air-conditioning systems, which harbor infected amoebae and produce microdroplets of water that are easily inhaled. Once in the lung, the bacteria are engulfed by macrophages by an unusual process called coiling phagocytosis (Figure 23-27A). *L. pneumophila* uses a type IV secretion system to inject effector proteins into the phagocyte that modulate the accumulation of phosphoinositides and the activity of proteins that regulate vesicular traffic, including SNARE proteins and Rab and Arf family small GTPases (discussed in Chapter 13). The effector proteins thereby prevent the phagosome from fusing with endosomes and promote its fusion with the endoplasmic reticulum, converting the phagosome into a compartment that resembles the rough endoplasmic reticulum (Figure 23-27B).

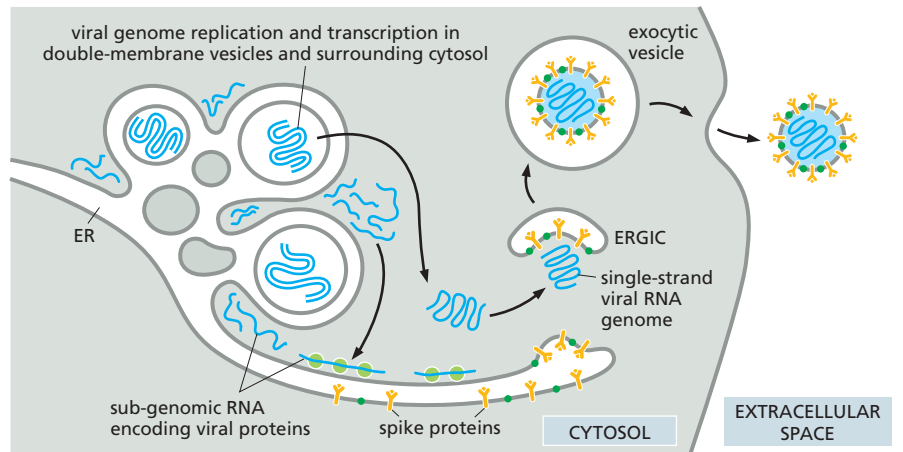
Viruses can also alter membrane traffic in the host cell. Many enveloped viruses, including the coronavirus SARS-CoV-2, make use of host-cell membranes to acquire their own envelope membrane. In the simplest cases, virally encoded glycoproteins are inserted into the endoplasmic reticulum membrane and follow

**Figure 23-27** *Legionella pneumophila* residing in a compartment with characteristics similar to those of the rough endoplasmic reticulum (ER).

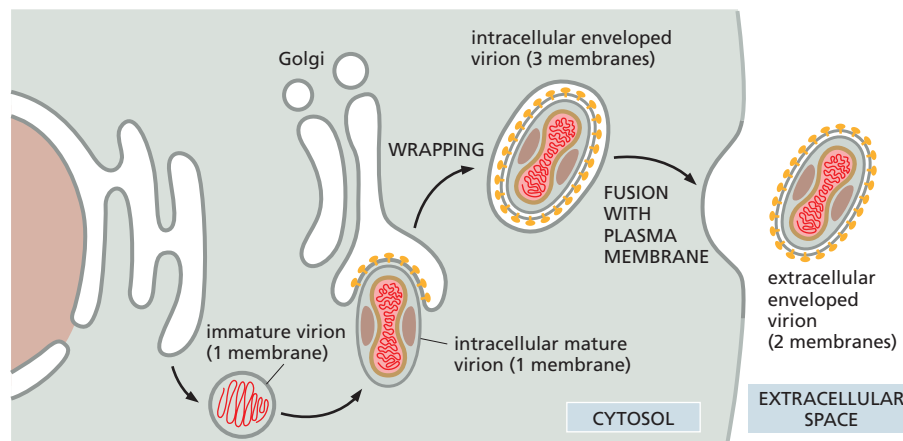
(A) Electron micrograph showing the unusual coiled structure that the *Legionella pneumophila* bacterium induces on the surface of a phagocyte during the invasion process. Some other pathogens, including the bacterium *Borrelia burgdorferi* (which causes Lyme disease), the eukaryotic pathogen *Leishmania*, and the yeast *Candida albicans*, can also invade cells using this type of coiling phagocytosis. (B) After invasion, *L. pneumophila* uses its type IV secretion system to secrete effector proteins that block phagosome–endosome fusion and phagosome maturation. It also secretes effector proteins that promote phosphoinositide conversion from PI(3)P to PI(4)P and the fusion of the phagosome with ER-derived vesicles, thereby converting the characteristics of the *Legionella*-containing vacuole from those of an endosome to those of the rough ER. See Figures 13-10 and 13-11 for a discussion of phosphoinositides and membrane targeting. (A, from M.A. Horwitz, *Cell* 36:27–33, 1984. With permission from Elsevier.)



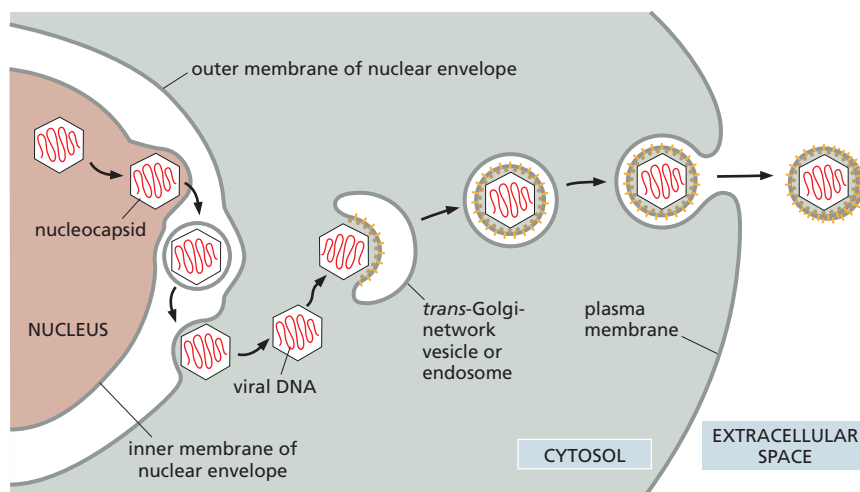
the secretory pathway through the Golgi apparatus to the plasma membrane; the viral capsid proteins and genome assemble into nucleocapsids, which acquire their envelope as they bud off from the plasma membrane (see Figure 23-12). The process for SARS-CoV-2 is more complicated, as this virus induces the formation of convoluted double-membrane vesicles derived from the endoplasmic reticulum, which house its genome replication factories (Figure 23-28A). These



(A) SARS-CoV-2



(B) Vaccinia virus



(C) Herpesvirus

**Figure 23-28 Complex strategies for viral envelope acquisition.**

(A) The coronavirus SARS-CoV-2 replicates its positive sense RNA genome in double membrane vesicles associated with the endoplasmic reticulum (ER). Viral genomes are released into the cytoplasm by a poorly understood mechanism that may involve passage through pores in the double membrane. Transcription of sub-genomic RNAs encoding viral structural proteins occurs in the surrounding cytosol. Translation of transmembrane viral proteins, including the spike protein, occurs on the ER. These proteins are then trafficked from the ER to the ER Golgi intermediate compartment (ERGIC), where the single-stranded viral RNA genome associates with viral proteins to drive formation of virus particles in the lumen of exocytic vesicles. When these vesicles fuse with the plasma membrane, viruses are released from the cell surface. (B) Vaccinia virus assembles in “replication factories” in the cytosol, far away from the plasma membrane. The immature virion, surrounded by a single membrane, then acquires two additional membranes from the Golgi apparatus by a poorly understood wrapping mechanism, to form the intracellular enveloped virion. After fusion of the outermost membrane with the host-cell plasma membrane, the extracellular enveloped virion is released from the host cell. (C) Herpesvirus nucleocapsids assemble in the nucleus and then bud through the inner nuclear membrane into the space between the inner and outer nuclear membranes, acquiring a membrane coat. The virus particles then apparently lose this coat when they fuse with the endoplasmic reticulum membrane to escape into the cytosol. Subsequently, the nucleocapsids travel through the Golgi apparatus, acquiring two new membrane coats in the process. The virus then buds from the cell surface with a single membrane when its outer membrane fuses with the plasma membrane.

may concentrate replication intermediates and sequester such intermediates from innate immune sensor molecules (see Chapter 24). Viral genomes are then released into the cytoplasm by an unknown mechanism and interact with viral structural proteins in the endoplasmic reticulum Golgi intermediate compartment (ERGIC), acquiring two new lipid bilayer membrane coats. Subsequent fusion of virus-containing exocytic vesicles with the plasma membrane leads to the release of viruses with a single membrane envelope. DNA viruses such as herpesviruses and vaccinia virus also alter membrane traffic and acquire their lipid envelopes in complex ways (**Figure 23–28B and C**).

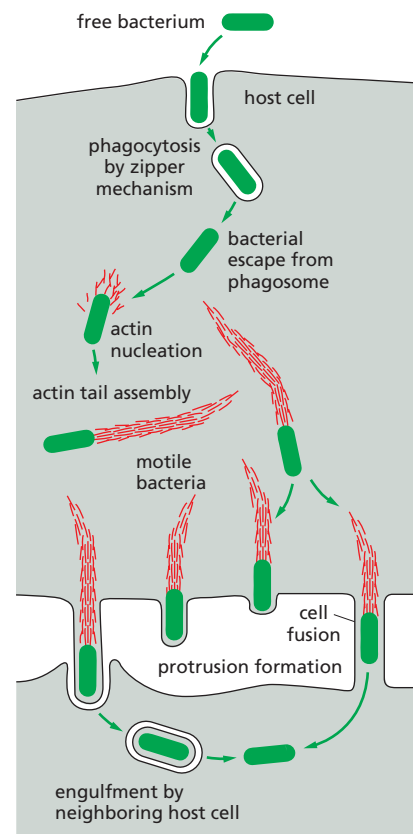
### Bacteria and Viruses Use the Host-Cell Cytoskeleton for Intracellular Movement

As mentioned earlier, many pathogens escape into the cytosol rather than remaining in a membrane-enclosed compartment. The cytosol of mammalian cells is extremely viscous, as it is crowded with protein complexes, organelles, and cytoskeletal filaments, all of which inhibit the diffusion of particles the size of a bacterium or a viral nucleocapsid. Thus, to reach a particular region of the host cell, a pathogen must be actively moved there. As with transport of intracellular organelles, pathogens generally use the host cell's cytoskeleton for their active movement.

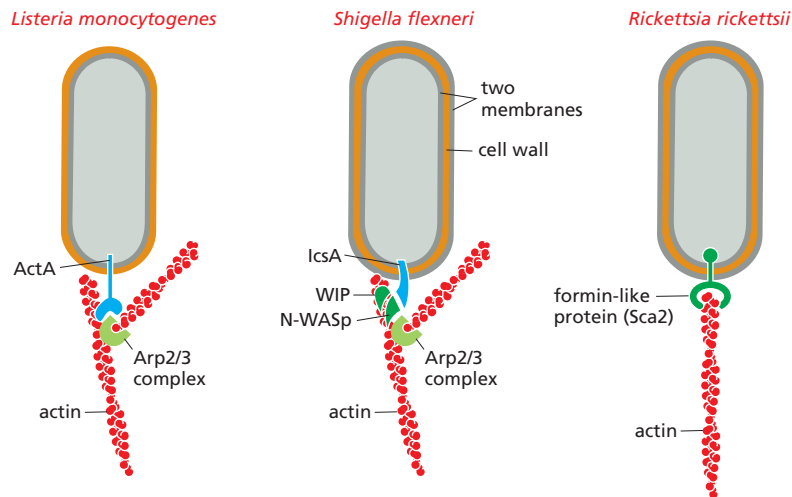
Several pathogens have adopted a remarkable mechanism that depends on actin polymerization for their movement. These include the human bacterial pathogens *Listeria monocytogenes*, *Shigella flexneri*, *Rickettsia rickettsii* (which causes Rocky Mountain spotted fever), and *Burkholderia pseudomallei* (which causes melioidosis, a disease characterized by severe respiratory symptoms), as well as Ebola virus and the insect virus baculovirus. All induce the nucleation and assembly of host-cell actin filaments at one pole of the microbe. The growing filaments generate force and push the pathogens through the cytosol at rates of up to 1  $\mu\text{m}/\text{min}$  (**Figure 23–29**). New filaments form at the rear of each pathogen and are left behind like a rocket trail as the microbe advances; the filaments depolymerize within a minute or so as they encounter depolymerizing factors in the cytosol. For *L. monocytogenes* and *S. flexneri*, the moving bacteria collide with the plasma membrane and move outward, inducing the formation of long, thin, host-cell protrusions with the bacteria at their tip. A neighboring cell often engulfs these projections, allowing the bacteria to enter the neighbor's cytoplasm without exposure to the extracellular environment, thereby avoiding antibodies produced by the host's adaptive immune system. For *B. pseudomallei*, movement and collision of the bacteria with the plasma membrane promote cell–cell fusion, which serves a similar purpose of immune avoidance while aiding continued bacterial replication.

The molecular mechanisms of pathogen-induced actin assembly differ for the different pathogens, suggesting that they evolved independently (**Figure 23–30**). *L. monocytogenes* and baculovirus produce proteins that directly bind to and activate the Arp2/3 complex to initiate the formation of an actin tail and movement (see Figure 16–17). *S. flexneri* produces an unrelated surface protein that binds to and activates the NPF N-WASp, which then activates the Arp2/3 complex. *Rickettsia* and *Burkholderia* species produce proteins that directly polymerize actin, for example by mimicking the function of host proteins such as formins (see Figure 16–13).

Many viral pathogens rely primarily on microtubule-dependent motor proteins, rather than actin polymerization, to move within the host-cell cytosol,

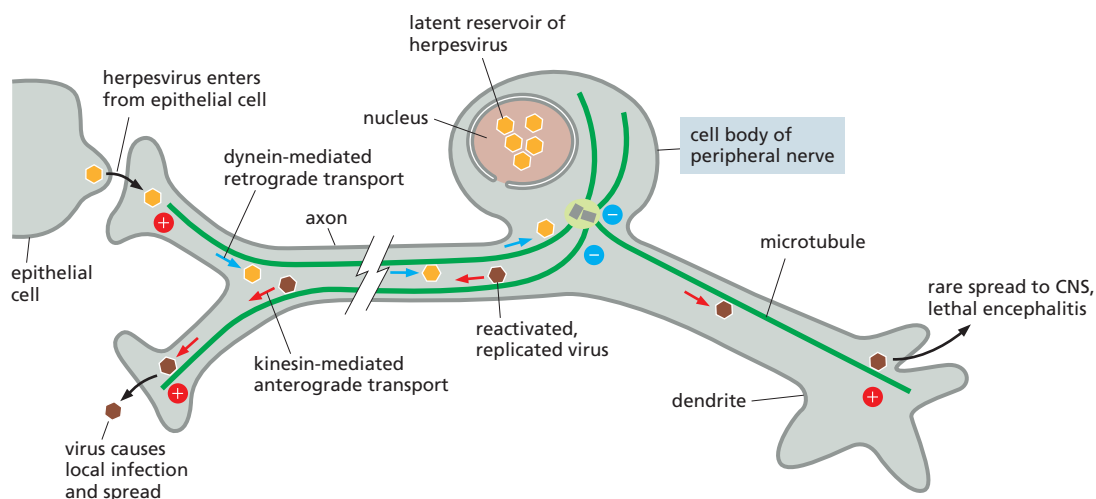


**Figure 23–29** The actin-based movement of bacterial pathogens within and between host cells. After invasion, bacterial pathogens such as *L. monocytogenes*, *S. flexneri*, *R. rickettsii*, and *B. pseudomallei* induce the assembly of actin-rich tails in the host-cell cytoplasm, which drives rapid bacterial movement. For most of these pathogens, the moving bacteria collide with the host-cell plasma membrane to form membrane-covered protrusions, which are engulfed by neighboring cells—spreading the infection from cell to cell. In contrast, for *B. pseudomallei*, collision with the plasma membrane promotes cell–cell fusion, creating a conduit through which bacteria can invade neighboring cells (**Movie 23.7**).

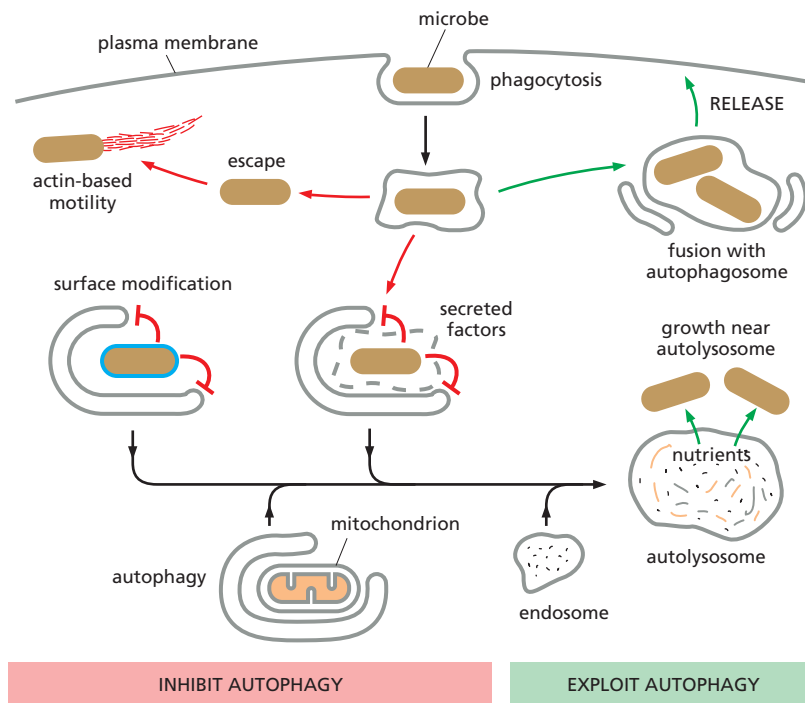


**Figure 23–30 Molecular mechanisms for actin nucleation by various bacterial pathogens.** *L. monocytogenes* and *S. flexneri* induce actin nucleation by recruiting and activating the host Arp2/3 complex (see Figure 16–12), although each uses a different recruitment strategy: *L. monocytogenes* expresses a surface protein, ActA, that directly binds to and activates the Arp2/3 complex; *S. flexneri* expresses a surface protein, IcsA (unrelated to ActA), that recruits the host NPF N-WASp, which in turn recruits the Arp2/3 complex, along with other host proteins, including WIP (WASp-interacting protein). *R. rickettsii* uses an entirely different strategy: it expresses a surface protein, Sca2, that directly nucleates actin polymerization by mimicking the activity of host formin proteins.

taking advantage of the inherent polarity of microtubules to enable directed long-range movements. Important examples are viruses that infect neurons, such as the *neurotropic alphaherpesviruses*, which include the virus that causes chickenpox (Figure 23–31). These virions enter sensory neurons at the tips of their axons and move by retrograde (“backward”) axonal transport along the axon toward the microtubule minus end. The transport is mediated by attachment of viral capsid proteins to the motor protein dynein. These viruses then establish either productive or latent infection in the nuclei of neurons of the peripheral nervous system. After replication or reactivation, virions are then carried by anterograde (“forward”) axonal transport along microtubules to the axon tips, with the transport being mediated by the attachment of a different viral capsid protein to a kinesin motor protein. Many other viruses associate with either dynein or kinesin motor proteins to move along microtubules at some stage in their replication, and some even alter the dynamic assembly and disassembly of microtubules. As microtubules serve as oriented tracks for vesicular transport in all cell types, not just neurons, it is not surprising that many viruses have independently evolved the ability to exploit them for their own transport.



**Figure 23–31 Microtubule-based movement of viruses.** Neurotropic alphaherpesviruses infect peripheral nerve cells from neighboring epithelial cells in the skin. Virus particles are then transported along microtubules in the nerve-cell axon toward the microtubule minus ends in the cell body (retrograde transport), a process that is driven by dynein motor proteins. Upon reaching the cell body, viruses can establish a latent reservoir in the nucleus. After reactivation (for example, following stress or immune suppression) or upon replication, virus particles are again transported along microtubules in the axon, this time toward the microtubule plus ends at the nerve terminal (anterograde transport), where they are released to enable local reinfection and spread. In rare instances, viruses can instead spread to the central nervous system (CNS), which can cause encephalitis.



**Figure 23–32 Microbial manipulation of autophagy.** Microbial pathogens have various mechanisms for avoiding antimicrobial autophagy (red lines), which include modifying their surface to avoid autophagy initiation or secreting factors and initiating actin-based motility to avoid enclosure in autophagosomes. Microbes can also exploit autophagy (green arrows), for example by recruiting and fusing with autophagosomes to gain nutrients and lipids or by growing near autolysosomes for nutrient acquisition.

### Many Microbes Manipulate Autophagy

Whether a microbe lives within the cytosol or inside a membrane-bound compartment, it must contend with the host cell's defenses, which include the *autophagy* pathway (see Figure 13–67). Autophagy is the process through which organelles or other cytoplasmic cargoes become surrounded by a double-membrane autophagosome that fuses with lysosomes to promote degradation (see Figure 13–71). It has recently become apparent that targeting of intracellular microbes for destruction by autophagy is an important pathway in the innate immune defense against infection (see Chapter 24). When this occurs, the process is called *antimicrobial autophagy*, or *xenophagy*. Not surprisingly, pathogens have evolved various strategies either to avoid antimicrobial autophagy or to manipulate autophagy pathways for their own benefit (Figure 23–32).

One strategy for avoiding antimicrobial autophagy is to deploy a protective shield that prevents detection of the microbe by the host cell. *Francisella tularensis*, which causes the zoonosis tularemia, or rabbit fever, uses this scheme. Once it escapes from the phagosome into the cytosol, it shields its surface by producing a variant of its outer membrane LPS that is not recognized by the host cell's autophagy initiation machinery. Other strategies to subvert autophagy, employed by *L. monocytogenes*, are to secrete bacterial enzymes and harness actin-based movement, both of which enable bacteria to avoid enclosure by autophagosomes. On the other hand, some pathogens take the opposite strategy of activating and then exploiting autophagy. *Coxiella burnetii*, which causes the zoonosis Q fever, replicates in a membrane-bound compartment that recruits and fuses with autophagosomes to deliver nutrients as well as lipids for compartment expansion. Poliovirus makes use of autophagy for the alternative effect of promoting virus trafficking to the plasma membrane and release from the host cell.

### Viruses Can Take Over the Metabolism of the Host Cell

Viruses use basic host-cell machinery for most aspects of their reproduction: they depend on host-cell ribosomes to produce their proteins, and many use host-cell DNA and RNA polymerases for their own replication and transcription. Many viruses encode proteins that modify the host transcription or translation apparatus to favor the synthesis of viral RNAs and proteins over those of the host

cell, shifting the synthetic capacity of the cell toward the production of new virus particles. Poliovirus, for example, encodes a protease that specifically cleaves the TATA-binding component of TFIID (see Figure 6–15), shutting off transcription of most of the host cell's protein-coding genes. Influenza virus produces a protein that blocks both the splicing and the polyadenylation of host-cell RNA transcripts, preventing their export into the cytosol (see Figure 6–39), but does not affect viral RNA transcripts.

Viruses also alter translation by the host. Translation initiation for most host-cell mRNAs depends on recognition of their 5' cap by translation initiation factors (see Figure 6–40). This initiation process is often inhibited during viral infection, so that the host-cell ribosomes can be used more efficiently for the synthesis of viral proteins. Some viral genomes encode endonucleases that cleave off the 5' cap from host-cell mRNAs; some go even further by using the liberated 5' caps as primers to synthesize viral mRNAs, a process called *cap snatching*. Several other viral RNA genomes encode proteases that cleave certain translation initiation factors; these viruses rely on 5' cap-independent translation of their own RNA, using internal ribosome entry sites (IRESs; see Figure 7–72).

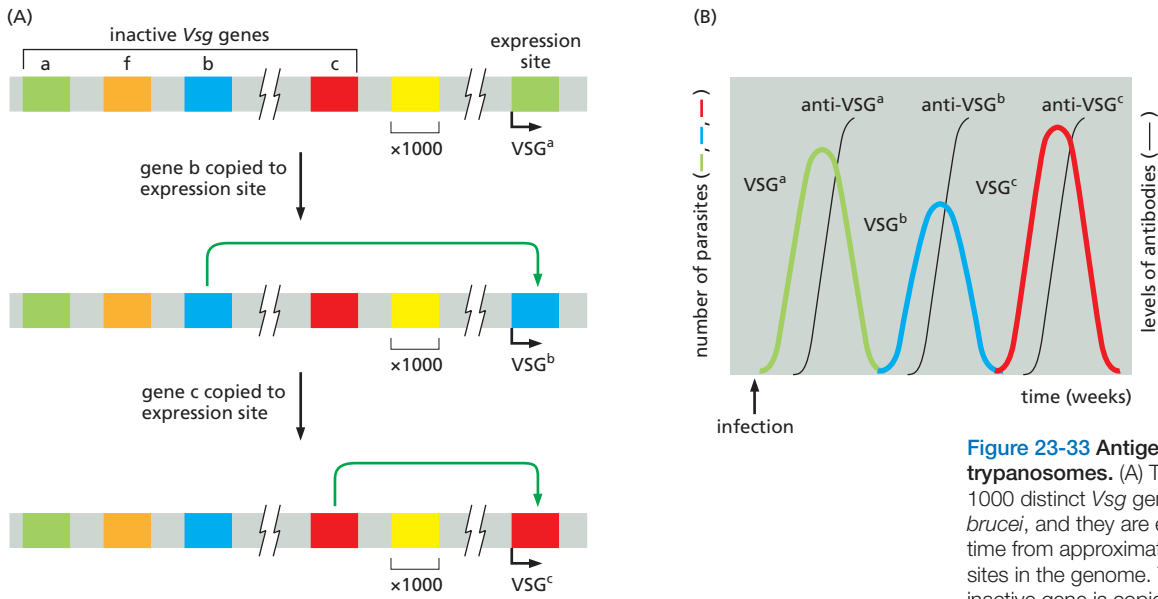
DNA viruses that replicate in the nucleus use host-cell DNA polymerase to replicate their genome. Because DNA polymerase is expressed at high levels only during S phase of the cell cycle, adenovirus has evolved a mechanism to drive the host cell into S phase, so that the cell produces large amounts of active DNA polymerase, which then replicates the viral genome. To accomplish this, the adenovirus genome also encodes proteins that inactivate both Rb (see Figure 17–59) and p53 (see Figure 17–60), two key suppressors of cell-cycle progression. As might be expected for any mechanism that encourages unregulated DNA replication, these viruses can promote, under some circumstances, the development of cancer.

Animal RNA viruses encode their own replication proteins because most animals lack polymerase enzymes that use RNA as a template (such as the RNA-dependent RNA polymerases used in RNA interference; see Chapter 7). For RNA viruses with a single-stranded genome, the replication strategy depends on whether the RNA is a positive [+] strand, which contains translatable information like mRNA does, or a complementary negative [–] strand. When the RNA genome is a positive [+] strand (such as for the coronavirus SARS-CoV-2), the incoming viral genome is translated to produce the viral RNA polymerase and viral proteins; the viral polymerase is then used to replicate the viral RNA and to generate mRNAs for the production of more viral proteins. For viruses with a negative [–] strand RNA genome (such as influenza and measles virus), an RNA polymerase enzyme is packaged as a structural protein of the incoming viral capsids.

Retroviruses such as HIV, which have a positive [+] strand RNA genome, are a special class of RNA virus because they carry with them a viral *reverse transcriptase* enzyme. After entry to the host cell, the reverse transcriptase uses the viral RNA genome as a template to synthesize a double-stranded DNA copy of the viral genome, which enters into the nucleus and integrates into the host cell's chromosomes (see Figure 5–61). It is later transcribed by the cell's DNA-dependent RNA polymerase to produce viral genomes and proteins.

### Pathogens Can Evolve Rapidly by Antigenic Variation

The complexity and specificity of the interplay between pathogens and their host cells might suggest that virulence would be difficult to acquire by random mutation. Yet, new pathogens are constantly emerging, and old pathogens are constantly changing in ways that make familiar infections more difficult to prevent or treat. Pathogens have two advantages that enable them to evolve rapidly. First, they replicate very quickly, providing a great deal of mutational variation for natural selection to work with. Whereas humans and chimpanzees have acquired a 2% difference in genome sequences over about 8 million years of divergent evolution, poliovirus manages a 2% change in its genome in 5 days—about the time it takes the virus to pass from the human mouth to the gut. Second, selective pressures act rapidly on this genetic variation. The host's adaptive immune



**Figure 23-33 Antigenic variation in trypanosomes.** (A) There are about 1000 distinct *Vsg* genes in *Trypanosoma brucei*, and they are expressed one at a time from approximately 20 expression sites in the genome. To be expressed, an inactive gene is copied and the copy is moved into an expression site through DNA recombination. Each *Vsg* gene encodes a different surface protein (antigen). These switching events allow the trypanosome to repeatedly change the surface antigen it expresses. (B) A person infected with trypanosomes expressing *VSG*<sup>a</sup> mounts an antibody response against this particular antigen, which clears most of the *VSG*<sup>a</sup>-expressing parasites. However, a few of the trypanosomes will have spontaneously switched to expression of *VSG*<sup>b</sup>, which can now proliferate until anti-*VSG*<sup>b</sup> antibodies are made. By that time, however, some parasites will have switched to *VSG*<sup>c</sup>, and so the cycle continues.

system and modern microbicidal drugs, both of which destroy pathogens that fail to change, are the main sources of these selective pressures.

An example of an adaptation to the selective pressure imposed by the adaptive immune system is the phenomenon of **antigenic variation**. An important adaptive immune response against many pathogens is the host's production of antibodies that recognize specific molecules (*antigens*) on the pathogen's surface (discussed in Chapter 24). Many pathogens have evolved mechanisms that change these antigens during the course of an infection, enabling them to stay one step ahead of the antibody response. Some eukaryotic parasites, for example, undergo programmed rearrangements of the genes encoding their surface antigens (Figure 23-33). A striking example occurs in *Trypanosoma brucei*, a protozoan parasite that causes African sleeping sickness and is spread by tsetse flies. (*T. brucei* is a relative of *T. cruzi*—see Figure 23-22—but it replicates extracellularly rather than intracellularly.) *T. brucei* is covered with a single type of glycoprotein, called *variant-specific glycoprotein* (VSG), which elicits in the host an antibody response that rapidly clears most of the parasites. The trypanosome genome, however, contains about 1000 different inactive *Vsg* genes (or pseudogenes), each encoding a VSG with a distinct amino acid sequence. Only one *Vsg* gene is expressed at any one time, from one of approximately 20 possible expression sites in the genome. Gene rearrangements that copy different inactive *Vsg* genes into expression sites repeatedly change the VSG protein displayed on the surface of the pathogen. In this way, a few trypanosomes with an altered VSG escape the initial antibody-mediated clearance, replicate, and cause the disease to recur, leading to a chronic cyclic infection.

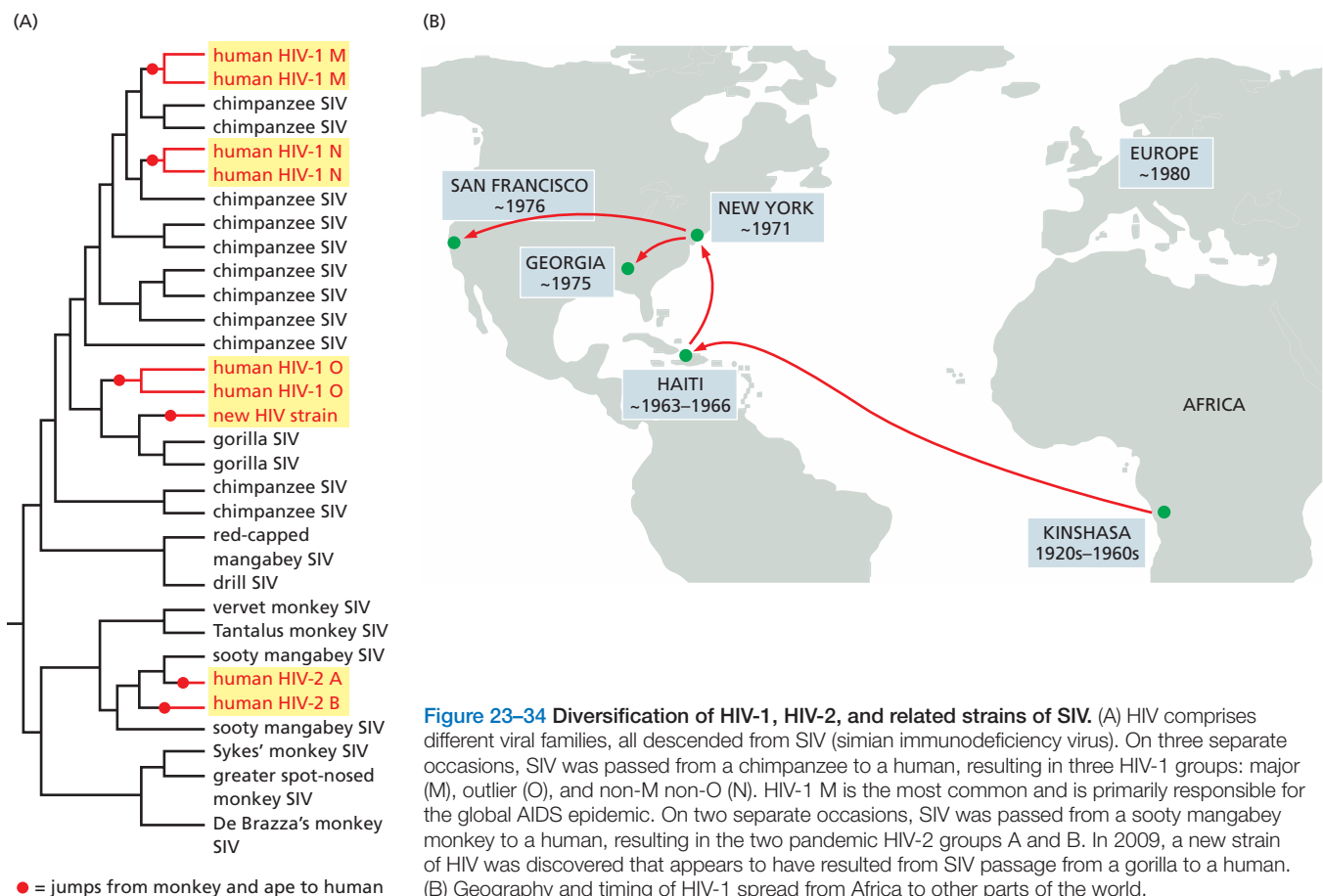
Bacterial pathogens can also rapidly change their surface antigens. Species of the genus *Neisseria* are champions at this. These Gram-negative cocci can cause sexually transmitted disease, in the case of *Neisseria gonorrhoeae*, or meningitis, in the case of *Neisseria meningitidis*. They undergo genetic recombination very similar to that just described for eukaryotic pathogens, which enables them to vary the pilin protein they use to attach to host cells. By inserting one of the multiple silent copies of variant *pilin* genes into a single expression locus, they can express many slightly different versions of the protein and repeatedly change the amino acid sequence over time. *Neisseria* bacteria are also extremely adept at taking up DNA from their environment by natural transformation and incorporating it into their genomes, further contributing to their extraordinary variability. The end result of this considerable variation is a plethora of different surface protein compositions with which to bewilder the host adaptive immune system. It is therefore not surprising that it has been difficult to develop an effective vaccine against *N. gonorrhoeae* infections, although there are now several that protect

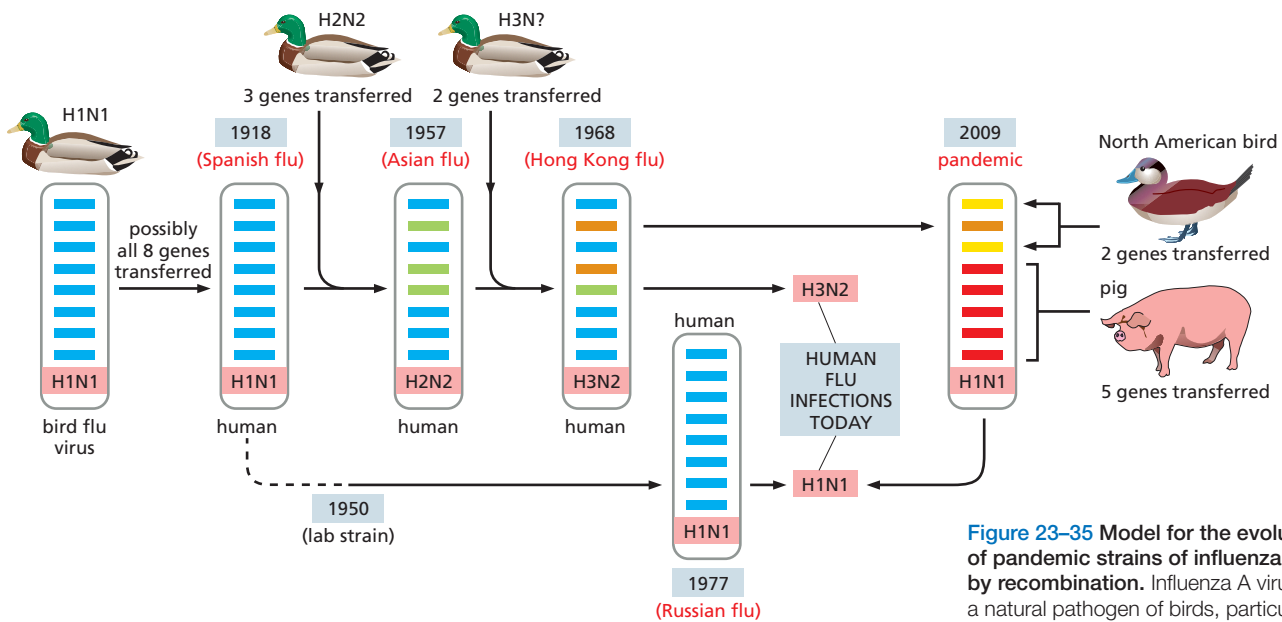
against *N. meningitidis*, due to the much smaller number of variants of its surface polysaccharide capsule.

### Error-prone Replication Dominates Viral Evolution

In contrast to the DNA rearrangements in bacteria and parasites, viruses rely on an error-prone replication mechanism for antigenic variation. Retroviral genomes, for example, acquire on average one point mutation every replication cycle, because the viral reverse transcriptase (see Figure 5-61) needed to produce DNA from the viral RNA genome lacks the proofreading activity of DNA polymerases. A typical, untreated HIV infection may eventually produce HIV genomes with every possible point mutation. By a process of mutation and selection within each host, most HIV viruses change over time—from a form that is most efficient at infecting macrophages to one more efficient at infecting T cells, as described earlier (see Figure 23-18). Similarly, once a patient is treated with an antiviral drug, the viral genome can quickly mutate and be selected for its resistance to the drug. Remarkably, only about one-third of the nucleotide positions in the coding sequence of the viral genome are invariant (because mutations at these positions would be detrimental to the virus), and nucleotide sequences in some parts of the genome, such as the *Env* gene (see Figure 7-66), can differ by as much as 30% from one HIV isolate to another. This extraordinary genomic plasticity greatly complicates attempts to develop vaccines against HIV.

The rapid evolution of HIV by error-prone replication has also led to the swift emergence and spread of new HIV strains. Nucleotide sequence comparisons between various strains of HIV and the very similar simian immunodeficiency virus (SIV) isolated from a variety of monkey species suggest that the most virulent type of HIV, HIV-1, may have jumped from primates to humans multiple independent times, starting as long ago as 1908 (Figure 23-34A). Sequence





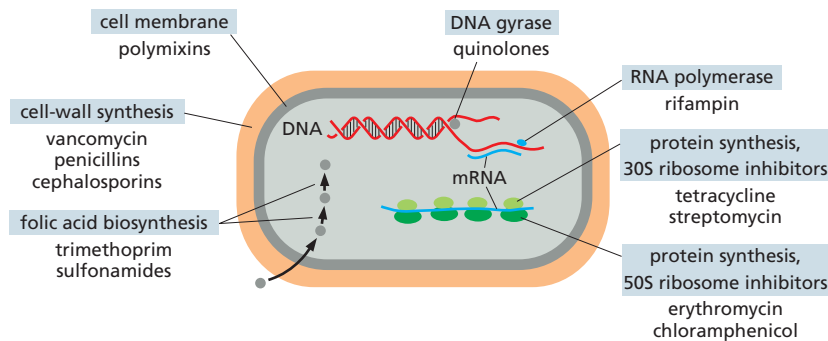
**Figure 23-35 Model for the evolution of pandemic strains of influenza virus by recombination.** Influenza A virus is a natural pathogen of birds, particularly waterfowl, and it is always present in wild bird populations. In 1918, a particularly virulent form of the virus crossed the species barrier from birds to humans and caused a devastating worldwide epidemic. This strain was designated H1N1, referring to the specific forms of its main antigens, hemagglutinin (H) and neuraminidase (N). Changes in the virus, rendering it less virulent, and the rise of adaptive immunity in the human population prevented the pandemic from continuing in subsequent seasons, although H1N1 influenza strains continued to cause serious disease every year in very young and very old people. In 1957, a new pandemic arose when three genes were replaced by equivalent genes from a different avian virus (green bars); the new strain (designated H2N2) was not effectively cleared by antibodies in people who had previously contracted only H1N1 forms of influenza. In 1968, another pandemic was triggered when two genes were replaced from yet another avian virus; the new virus was designated H3N2. In 1977, there was a resurgence of H1N1 influenza, which had previously been almost completely replaced by the N2 strains. Molecular sequence information suggests that this minor pandemic may have been caused by an accidental release of an influenza strain that had been held in a laboratory since about 1950 or by the use of this strain in a vaccine study. In 2009, a new H1N1 swine virus emerged that had derived five genes from pig influenza viruses, two from avian influenza viruses, and one from a human influenza virus. As indicated, most human influenza today is caused by H1N1 and H3N2 strains.

comparisons between human HIV strains has further enabled a reconstruction of the evolutionary and geographic history of the current pandemic of HIV-1 group M (major). The pandemic strain is thought to have originated in Kinshasa, Democratic Republic of the Congo, in the 1920s. Virus spread increased in Africa in the 1960s due to changing sexual behaviors and emerging transportation networks (Figure 23-34B). It then crossed the Atlantic to Haiti in the late 1960s and spread to New York and other locations in the United States in the 1970s, eventually giving rise to a worldwide pandemic.

An important exception to the rule that error-prone replication dominates viral evolution is the influenza viruses. Although they accumulate point mutations as they replicate, they differ from other viruses in that their genome consists of several (usually eight) strands of RNA, each of which codes for different proteins. When two strains of influenza infect the same host, the RNA strands of the two strains can reassort to form a new type of influenza virus. In normal years, influenza is a mild disease in healthy adults, although it can be life-threatening in the very young and very old. Different influenza strains infect fowl such as ducks and chickens, but only a subset of these strains can infect humans, and transmission from fowl to humans is rare. In 1918, however, a particularly virulent variant of avian influenza crossed the species barrier to infect humans, triggering the catastrophic pandemic of 1918 called the Spanish flu, which killed 20–50 million people worldwide. Subsequent influenza pandemics have been triggered by genome reassortment, in which a new RNA segment from an avian form of the virus replaced one or more of the viral RNA segments from the human form (Figure 23-35). In 2009, a new H1N1 swine virus emerged that derived genes from pig, avian, and human influenza viruses. Such recombination events allowed the new virus to replicate rapidly and spread through an immunologically naïve human population. Generally, within 2 or 3 years, the human population develops immunity to a new recombinant strain of virus, and the infection rate drops to a steady-state level. Because the recombination events are unpredictable, it is not possible to know when the next influenza pandemic will occur or how severe it might be.

### Drug-resistant Pathogens Are a Growing Problem

The development of drugs that cure rather than prevent infections has had a major impact on human health. **Antibiotics**, which are either bactericidal (they kill bacteria) or bacteriostatic (they inhibit bacterial growth without killing), are the most successful class of such drugs. Penicillin was one of the first antibiotics

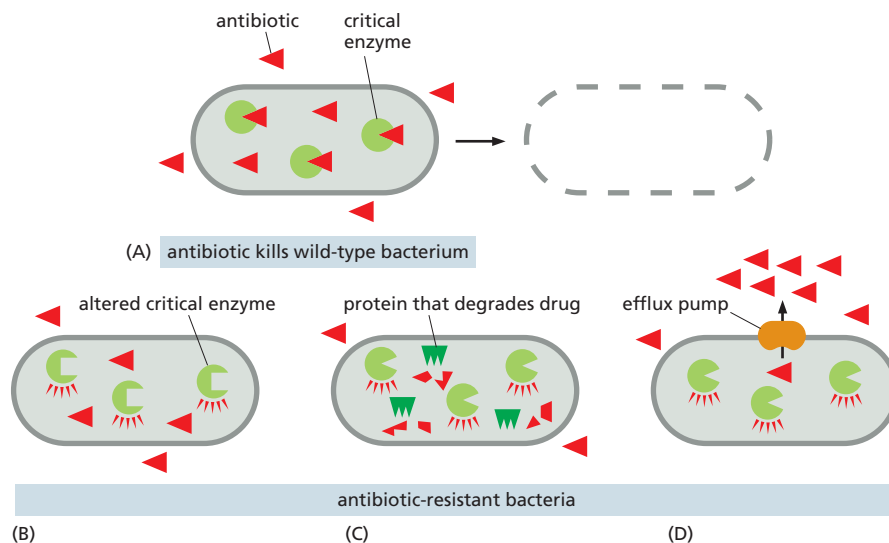


**Figure 23–36 Antibiotic targets.** Although there are many antibiotics in clinical use, they have a narrow range of targets, which are highlighted in *blue*. A few representative antibiotics in each class are listed. Nearly all antibiotics used to treat human infections fall into one of these categories. The vast majority inhibit either bacterial protein synthesis or bacterial cell-wall synthesis. The illustration example is a Gram-positive bacterium.

used to treat infections in humans, just in time to prevent tens of thousands of deaths from infected battlefield wounds in World War II. Because bacteria (see Figure 1–9) are not closely related evolutionarily to the eukaryotes they infect, much of their basic machinery for cell-wall synthesis, DNA replication and transcription, RNA translation, and metabolism differs from that of their host. These differences enable us to develop antibacterial drugs that exhibit *selective toxicity*, in that they specifically inhibit these processes in bacteria without disrupting them in the host. Most of the antibiotics that we use to treat bacterial infections are small molecules that inhibit macromolecular synthesis in bacteria by targeting bacterial enzymes that either are distinct from their eukaryotic counterparts or are involved in pathways such as cell-wall biosynthesis that are absent in animals (Figure 23–36; see also Table 6–4).

However, bacteria continually evolve and strains resistant to antibiotics rapidly develop, often within a few years of the introduction of a new drug. Similar drug resistance also arises rapidly when treating viral infections with antiviral drugs. The virus population in an HIV-infected person treated with the reverse transcriptase inhibitor azidothymidine (AZT), for example, will acquire complete resistance to the drug within a few months. The current protocol for treatment of HIV infections involves the simultaneous use of three drugs, which helps to minimize the acquisition of resistance for any one of them. Even eukaryotic pathogens rapidly evolve resistance. The malaria parasite *Plasmodium falciparum* is now generally resistant to the heavily used drug chloroquine, which was introduced in the 1930s, and resistance to the newer drug artemisinin is emerging.

There are three general strategies by which a pathogen can develop drug resistance: (1) it can alter the molecular target of the drug so that it is no longer sensitive to the drug; (2) it can produce an enzyme that modifies or destroys the drug; or (3) it can prevent the drug's access to the drug target by, for example, actively pumping the drug out of the pathogen (Figure 23–37).



**Figure 23–37 Three general mechanisms of antibiotic resistance.** (A) A nonresistant wild-type bacterial cell bathed in a drug (red triangles) that binds to and inhibits an essential enzyme (light green) will be killed because of enzyme inhibition. (B) A bacterium that has altered the drug's target enzyme so that the drug no longer binds to the enzyme will survive and proliferate. In many cases, a single point mutation in the gene encoding the target protein can generate resistance. (C) A bacterium that expresses an enzyme (dark green) that either degrades or covalently modifies the drug will survive and proliferate. Some resistant bacteria, for example, make  $\beta$ -lactamase enzymes, which cleave penicillin and similar molecules. (D) A bacterium that expresses or up-regulates an efflux pump that ejects the drug from the bacterial cytoplasm (using energy derived from either ATP hydrolysis or the electrochemical gradient across the bacterial plasma membrane) will survive and proliferate. Some efflux pumps, such as the TetR efflux pump, are specific for a single drug (in this case, tetracycline), whereas others, called multidrug resistance (MDR) efflux pumps, are capable of exporting a wide variety of structurally dissimilar drugs. Up-regulation of an MDR pump can render a bacterium resistant to a very large number of different antibiotics in a single step.

Once a pathogen has chanced upon an effective drug-resistance strategy, the newly acquired or mutated genes that confer the resistance are frequently spread throughout the pathogen population by horizontal gene transfer. They may even spread between pathogens of different species. The highly effective but expensive antibiotic *vancomycin*, for example, is used as a treatment of last resort for many severe, hospital-acquired, Gram-positive bacterial infections that are resistant to most other known antibiotics. Vancomycin prevents one step in bacterial cell-wall synthesis—the cross-linking of peptidoglycan chains in the bacterial cell wall (see Figure 23–3B). Resistance can arise if the bacterium synthesizes a cell wall using different subunits that do not bind vancomycin. The most devastating form of vancomycin resistance depends on the acquisition of a transposon (see Figure 5–58) containing seven genes, the products of which work together to sense the presence of vancomycin, shut down the normal pathway for bacterial cell-wall synthesis, and produce a different type of cell wall.

Drug-resistance genes acquired by horizontal transfer frequently come from environmental microbial reservoirs. Nearly all antibiotics used to treat bacterial infections today are based on natural products produced by fungi or bacteria. Penicillin, for example, is made by the mold *Penicillium*, and more than 50% of the antibiotics currently used in the clinic are made by Gram-positive bacteria of the genus *Streptomyces*, which reside in the soil. It is believed that microorganisms produce antimicrobial compounds, many of which have probably existed on Earth for hundreds of millions of years, as weapons in their competition with other microorganisms in the environment. Surveys of bacteria taken from soil samples that have never been exposed to antibiotic drugs used in modern medicine reveal that there are bacteria already resistant to about seven or eight of the antibiotics widely used in clinical practice. When pathogenic microorganisms are faced with the selective pressure provided by antibiotic treatments, they can apparently draw upon the immense source of genetic material in environmental microbial reservoirs to acquire resistance.

Like most other aspects of infectious disease, human behavior has exacerbated the problem of drug resistance. Many patients take antibacterial antibiotics for symptoms that are typically caused by viruses (flu-like illnesses, colds, and sore throats), and these drugs have no effects. Persistent and chronic misuse of antibiotics can eventually result in antibiotic-resistant microbes, which can then transfer the resistance to pathogens. Antibiotics are also misused in the livestock industry, where they are commonly employed as food additives to promote the growth and health of farm animals. An antibiotic closely related to vancomycin was commonly added to cattle feed in Europe; the resulting resistance in the microbiota of these animals is widely believed to be one of the original sources for vancomycin-resistant bacteria that now threaten the lives of hospitalized patients.

## Summary

*All pathogens share the ability to interact with host cells in diverse ways that promote pathogen replication and spread. Pathogens often colonize the host by adhering to or invading the epithelial surfaces that line the respiratory, gastrointestinal, and urinary tracts, as well as the other body surfaces in direct contact with the environment. Intracellular pathogens, including all viruses and many bacteria and protozoa, invade host cells by one of several mechanisms. Viruses rely largely on receptor-mediated endocytosis, whereas bacteria exploit cell adhesion and phagocytic pathways; in both cases, the host cell provides the machinery and energy for the invasion. Protozoa, by contrast, employ unique invasion strategies that usually require significant energy expenditure on the part of the invader. Once inside, intracellular pathogens seek out a cell compartment that is favorable for their survival and replication, frequently altering host membrane traffic, exploiting the host-cell cytoskeleton for intracellular movement, and manipulating autophagy. Pathogens evolve rapidly, so that new infectious diseases frequently emerge, and old pathogens acquire new ways to evade our attempts at prevention, treatment, and eradication.*

*Because the acquisition of drug resistance is almost inevitable, it is crucial that we take measures to delay the onset of resistance and develop innovative new drug treatments.*

## THE HUMAN MICROBIOTA

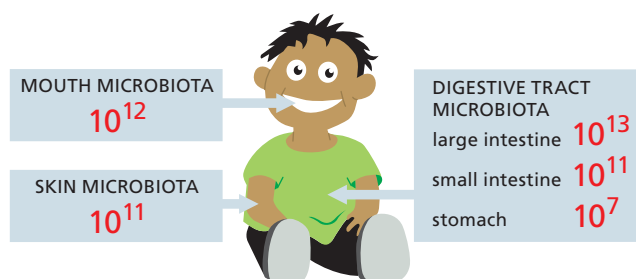
Although pathogens can cause disease in otherwise healthy humans, it is now appreciated that our bodies are colonized by many microbes that normally cause no harm. These microbes compose the so-called microbiota. Recent estimates suggest that the human body contains about  $3 \times 10^{13}$  human cells, as well as approximately  $4 \times 10^{13}$  resident microbial cells. As detailed below, we now appreciate that the microbiota is a complex microbial community that makes important contributions to human biology.

### The Human Microbiota Is a Complex Ecological System

The human microbiota is usually confined to specific locations of the body (Figure 23–38), including the skin, mouth, digestive tract, and vagina, with distinct communities of species inhabiting each body part. Of the resident microbes, bacterial cells make up the vast majority by sheer numbers (>99%), whereas there are smaller numbers of archaeal, fungal, and protozoan cells. The concentrations and total numbers of microbial cells differ vastly between body locations (Figure 23–38), with the large intestine containing by far the most microbial residents.

The microbiota of an individual person consists of thousands of different microbial species. The species composition varies considerably between individual humans, even between close relatives or identical twins. Different body sites also have different degrees of species diversity. The digestive tract, for example, contains between 500 and 1000 species, most belonging to only a few phyla of bacteria. The microbiota of the digestive tract also provides an interesting illustration of how the diversity and composition of microbial species can change over time, from birth through adulthood. The digestive tract of human infants is colonized by environmental microbes during birth, and the mode of birth (vaginal delivery versus cesarean section) can influence species composition. During the first year of life, the microbiota consists of fewer species that vary considerably over time and between individuals, whereas species diversity increases and composition stabilizes at 1–2 years of age. Thereafter, the microbiota of an individual is generally consistent over time but is influenced by a variety of factors, including age, pregnancy, diet, health status, hygiene, and antibiotic use.

To appreciate the contributions of the microbiota to human biology, it is helpful to consider the various ecological terms that are used to classify relationships between microbes and their host. For the pathogens discussed earlier in this chapter, the microbe benefits to the detriment of the host, a situation referred to as **parasitism**. In contrast, most constituents of the microbiota exhibit less sinister ecological relationships with the host. Some exhibit **commensalism**, which describes the circumstance in which the microbe benefits but has no known beneficial or harmful effects on the host. In **mutualism**, both the microbe and host benefit. It is sometimes difficult to draw a line between these categories.



**Figure 23–38** Sites in the human body that harbor the microbiota include the skin, mouth, and digestive tract. An estimate of the total number of microbes in each location is indicated.

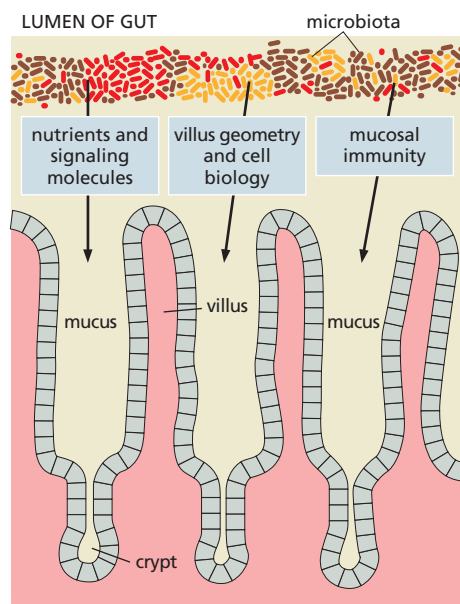
Some constituents of the microbiota that exhibit commensalism or mutualism under normal circumstances can also act as parasites or *opportunistic pathogens* that can cause disease if our immune systems are weakened or if they gain access to a normally sterile part of the body.

### The Microbiota Influences Our Development and Health

It is increasingly recognized that many constituents of the human microbiota exhibit a mutualistic relationship with our bodies by supporting metabolism, development, and immunity (Figure 23–39). The anaerobic bacteria that inhabit our intestines, for example, gain shelter and a nutrient supply but also contribute to the digestion of our food and produce important metabolites. One reason for the beneficial influence of the microbiota on our metabolism is that the combined genomes of the various microbial species, called the **microbiome**, contain 100 times more genes than the number in the human genome itself. A consequence of this genomic diversity is that the microbiota expands the range of biochemical activities available to humans by producing small-molecule metabolites of different chemical composition or in greater abundance than that produced by our own cells. An example is the production of abundant short-chain fatty acids by the microbiota in the digestive tract, which may contribute to metabolism and may also signal to our cells to influence human physiology.

The microbiota is also important for normal development, most notably of the epithelial tissue and immune system in the digestive tract. Studies comparing mice with and without a gastrointestinal microbiota (so-called *germ-free* mice) show that the microbiota affects properties of the intestinal lining including villus geometry, stem-cell proliferation, blood vessel density, and mucus thickness (Figure 23–39). The *mucosal immune system* that is associated with the mucus-containing surface of the intestinal epithelium is also strongly influenced by the microbiota. The mucosal immune system must be fine-tuned to be non-responsive to beneficial microbes in the microbiota, yet responsive to pathogens and microbes that inappropriately penetrate into the epithelial layer. The microbiota plays an important role in the development of lymphoid tissues and in the appropriate differentiation of various immune cell types to control the overall number, species composition, and location of microbes in the digestive tract (Figure 23–39).

There is increasing evidence that an imbalance in the community of microbes that constitute the microbiota, referred to as *dysbiosis*, is correlated with various human diseases. These include autoimmune and allergic diseases, obesity,



**Figure 23–39 Influences of the microbiota on metabolism and development.** The microbiota (not drawn exactly to scale) produces nutrients as well as signaling molecules that affect host-cell biology and physiology and also influences the development of the intestinal epithelium and the mucosal immune system.

inflammatory bowel disease, and diabetes, and the list is expanding rapidly. A common example of a disease caused by dysbiosis is colitis, which occurs in response to antibiotic treatment that kills off beneficial microbes in the digestive tract, resulting in overgrowth of the bacterial pathogen *Clostridium difficile*. Although first-line treatment for colitis is an additional course of antibiotics, recurrent colitis has been successfully treated by the transfer of the microbiota from a healthy individual to someone suffering from the disease.

As research intensifies into the influences of the microbiota on human health and disease, a future challenge will be to shift scientific inquiry from investigation of correlations and associations between the presence of certain microbes and health status to investigation of whether there are causal relationships between the composition of the microbiota and its molecular products and health or disease. An impediment to making this shift is the sheer number of variables that must be considered when studying a community consisting of hundreds of microbial species that produce thousands of proteins and small molecules. One path forward may be to search for clear instances in which individual microbes or molecules exert strong effects on human physiology. As microbiomes with defined composition are developed and genetic tools are improved, we may begin to understand how individual microbial species or even individual microbial genes influence human development and health, and we may be able to harness the microbiome to treat human disease.

## Summary

*Humans are colonized by a community of microbes, collectively called the microbiota, that consists of roughly the same number of microbial cells as there are human cells in the body and is located primarily on the skin, in the mouth, and in the digestive tract. The microbiota as a whole is thought to exhibit a mutualistic ecological relationship with the human host, meaning that both microbes and humans benefit from their interactions. The microbiota produces nutrients that aid in metabolism, as well as molecules that influence tissue and immune system development and function. However, it has been difficult to establish causal relationships between individual microbes or their molecular products and human health. Knowledge of the microbiota is expanding rapidly, and advances promise to enhance our understanding of human biology and result in new disease treatments.*

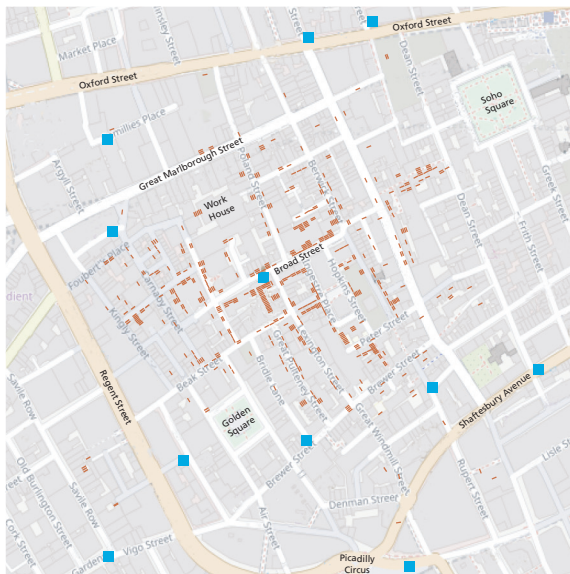
## PROBLEMS

Which statements are true? Explain why or why not.

- 23–1** Pathogens must enter host cells to cause disease.
- 23–2** Viruses replicate their genomes in the nucleus of the host cell.
- 23–3** You should not take antibiotics for diseases caused by viruses.
- 23–4** Our adult bodies harbor about 10 times more resident microbial cells than human cells.
- 23–5** The microbiomes from healthy humans are all very similar.

Discuss the following problems.

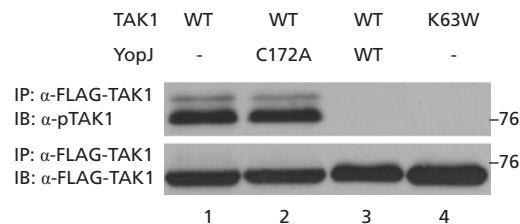
- 23–6** In order to survive and multiply, a successful pathogen must accomplish five tasks. Name them.
- 23–7** What are the three general mechanisms for horizontal gene transfer?
- 23–8** John Snow is widely regarded as the father of modern epidemiology. Most famously, he investigated an outbreak of cholera in London in 1854 that killed more than 600 people before it was finished. Snow recorded where the victims lived and plotted the data on a map, along with the locations of the water pumps that served as the source of water for the public (Figure Q23–1). He concluded that the disease was most likely spread in the water, although he could find nothing suspicious-looking



**Figure Q23–1** A map of where the victims of the 1854 cholera outbreak lived, superimposed on a modern map of the area (Problem 23–8). The locations of the victims' houses are indicated by the small red rectangles. Stacks of rectangles indicate multiple cases occurring in the same house. Public water pumps are shown as blue squares. (Adapted from Wellcome Library, London/Google Maps.)

in it. His conclusion ran counter to the then-current belief that cholera was from “miasmas” in bad air. Very few believed his theory during the next 50 years, with the “bad air” theory persisting until at least 1901. What do you suppose Snow saw in the data that led him to his conclusion? Why do you think most scientists remained skeptical for so long?

**23–9** The Gram-negative bacterium *Yersinia pestis*, the causative agent of the plague, is extremely virulent. Upon infection, *Y. pestis* injects a set of effector proteins into macrophages that suppresses their phagocytic behavior and also interferes with their innate immune responses. One of the effector proteins, YopJ, acetylates serines and threonines on various MAP kinases, including the MAP kinase kinase kinase (MAPKKK) TAK1, which controls a key signaling step in the innate immune response pathway. To determine how YopJ interferes with TAK1, you transfect human cells with catalytically active YopJ (YopJ<sup>WT</sup>) or inactive YopJ (YopJ<sup>C172A</sup>) and with FLAG-tagged active TAK1 (TAK1<sup>WT</sup>) or inactive TAK1 (TAK1<sup>K63W</sup>) and assay for total TAK1 and for phosphorylated TAK1, using antibodies against the FLAG tag or against phosphorylated TAK1 (Figure Q23–2). How does YopJ block the TAK1 signaling pathway? How do you suppose the serine/threonine acetylase activity of YopJ might interfere with TAK1 activation?



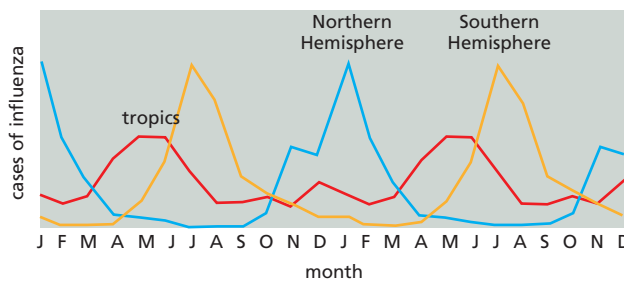
**Figure Q23–2** Effects of YopJ on TAK1 phosphorylation (Problem 23–9). TAK1 was immunoprecipitated (IP) using antibodies against the FLAG tag (α-FLAG-TAK1). Total TAK1 in the immunoprecipitation was assayed by immunoblot (IB) using the same antibody. Phosphorylated TAK1 was assayed by IB using antibodies specific for phospho-TAK1 (α-pTAK1). A marker of protein molecular mass is shown at right in kilodaltons. (From N. Paquette et al., *Proc. Natl. Acad. Sci. USA* 109:12710–12715, 2012. With permission from National Academy of Sciences.)

**23–10** The intracellular bacterial pathogen *Salmonella enterica* serovar Typhimurium, which causes gastroenteritis, injects effector proteins to promote its invasion into nonphagocytic host cells by the trigger mechanism. *S. enterica* serovar Typhimurium first stimulates membrane ruffling to promote invasion, and then suppresses membrane ruffling once invasion is complete. This behavior is mediated in part by injection of two effector proteins: SopE, which promotes membrane ruffling and invasion, and SptP, which blocks the effects of SopE. Both effector proteins target the monomeric GTPase, Rac, which in its active form promotes membrane ruffling. How do you

suppose SopE and SptP affect Rac activity? How do you suppose the effects of SopE and SptP are staggered in time if they are injected simultaneously?

**23-11** Several negative-strand viruses carry their genome as a set of discrete RNA segments. Examples include influenza virus (eight segments), Rift Valley fever virus (three segments), hantavirus (three segments), and Lassa virus (two segments), to name a few. Why does segmentation of the genome provide a strong evolutionary advantage for these viruses?

**23-12** Influenza epidemics account for 250,000–500,000 deaths globally each year. These epidemics are markedly seasonal, occurring in temperate climates in the Northern and Southern Hemispheres during their respective winters. By contrast, in the tropics, there is significant influenza activity year round, with a peak in the rainy season (Figure Q23-3). Can you suggest some possible explanations for the patterns of influenza epidemics in temperate zones and the tropics?



**Figure Q23-3** Seasonal patterns of influenza epidemics (Problem 23-12). Cases of influenza at different times of the year are shown for the Northern Hemisphere (blue), the Southern Hemisphere (orange), and the tropics (red).

**23-13** Avian influenza viruses readily infect birds but are transmitted to humans very rarely. Similarly, human influenza viruses spread readily to other humans but have never been detected in birds. The key to this specificity lies in the viral capsid protein, hemagglutinin, which binds to sialic acid residues on cell-surface glycoproteins, triggering virus entry into the cell (Movie 23.8). Hemagglutinin on human viruses recognizes sialic acid in a 2,6 linkage with galactose, whereas avian hemagglutinin recognizes sialic acid in a 2,3 linkage with galactose. Humans make carbohydrate chains that have only the 2,6 linkage between sialic acid and galactose; birds make only the 2,3 linkage; but pigs make carbohydrate chains with both linkages. How does this situation make pigs ideal hosts for generating new strains of human influenza viruses?

**23-14** The majority of antibiotics used in the clinic are made as natural products by bacteria. Why do you suppose bacteria make the very agents we use to kill them?

**23-15** In the early days of penicillin research, it was discovered that bacteria in the air could destroy the penicillin, a big problem for large-scale production of the drug. How do you suppose this occurs?

**23-16** When the Oxford team of Ernst Chain and Norman Heatley had laboriously collected their first two grams of penicillin (probably no more than 2% pure!), Chain injected two normal mice with 1 g each of this preparation and waited to see what would happen. The mice survived with no apparent ill effects. His boss, Howard Florey, was furious at what he saw as a waste of good antibiotic. Why was this experiment important?

## REFERENCES

### General

- Acheson NH (2021) Fundamentals of Molecular Virology, 3rd ed. Hoboken, NJ: Wiley.
- Cossart P, Roy CR & Sansonetti P (eds.) (2019) Bacteria and Intracellularly. Washington, DC: ASM Press.
- Fauci A & Morens DM (2012) The perpetual challenge of infectious diseases. *N. Engl. J. Med.* 366, 454–461.
- Gunn A & Pitt SJ (2012) Parasitology: An Integrated Approach. Hoboken, NJ: Wiley.
- Heitman J, Filler SG, Edwards JE & Mitchell AP (2006) Molecular Principles of Fungal Pathogenesis. Washington, DC: ASM Press.
- Wilson BA, Winkler ME & Ho BT (2019) Bacterial Pathogenesis: A Molecular Approach, 4th ed. Washington, DC: ASM Press.

### Introduction to Pathogens

- Baltimore D (1971) Expression of animal virus genomes. *Bacteriol. Rev.* 35, 235–241.
- Coffin JM (2021) 50th anniversary of the discovery of reverse transcriptase. *Mol. Biol. Cell* 32, 91–97.
- Cowman AF, Healer J, Marapana D & Marsh K (2016) Malaria: biology and disease. *Cell* 167, 610–624.
- Crick FH & Watson JD (1956) Structure of small viruses. *Nature* 177, 473–475.

- Faruque SM & Mekalanos JJ (2012) Phage-bacterial interactions in the evolution of toxigenic *Vibrio cholerae*. *Virulence* 3, 556–565.
- Galán JE & Gabriel Waksman G (2018) Protein-injection machines in bacteria. *Cell* 172, 1306–1318.
- Rappleye CA & Goldman WE (2006) Defining virulence genes in the dimorphic fungi. *Annu. Rev. Microbiol.* 60, 281–303.
- Soucy S, Huang J & Gogarten J (2015) Horizontal gene transfer: building the web of life. *Nat. Rev. Genet.* 16, 472–482.
- Young JAT & Collier RJ (2007) Anthrax toxin: receptor binding, internalization, pore formation, and translocation. *Annu. Rev. Biochem.* 76, 243–265.

### Cell Biology of Pathogen Infection

- Asrat S, de Jesús DA, Hempstead AD . . . Isberg RR (2014) Bacterial pathogen manipulation of host membrane trafficking. *Annu. Rev. Cell Dev. Biol.* 30, 79–109.
- Davies J & Davies D (2010) Origins and evolution of antibiotic resistance. *Microbiol. Mol. Biol. Rev.* 74, 417–433.
- Faria NR, Rambaut A, Suchard MA . . . Lemey P (2014) HIV epidemiology. The early spread and epidemic ignition of HIV-1 in human populations. *Science* 346, 56–61.
- Forsberg KJ, Reyes A, Wang B . . . Dantas G (2012) The shared antibiotic resistome of soil bacteria and human pathogens. *Science* 337, 1107–1111.

- Goldberg DE, Siliciano RF & Jacobs WR, Jr (2012) Outwitting evolution: fighting drug-resistant TB, malaria, and HIV. *Cell* 148, 1271–1283.
- Grove J & Marsh M (2011) The cell biology of receptor-mediated virus entry. *J. Cell Biol.* 195, 1071–1082.
- Hartenian E, Nandakumar D, Lari A . . . Glaunsinger BA (2020) The molecular virology of coronaviruses. *J. Biol. Chem.* 295, 12910–12934.
- Hernandez-Gonzalez M, Larocque G & Way M (2021) Viral use and subversion of membrane organization and trafficking. *J. Cell Sci.* 134, jcs252676.
- Huang J & Brumell J (2014) Bacteria–autophagy interplay: a battle for survival. *Nat. Rev. Microbiol.* 12, 101–114.
- Mengaud J, Ohayon H, Gounon P . . . Cossart P (1996) E-cadherin is the receptor for internalin, a surface protein required for entry of *L. monocytogenes* into epithelial cells. *Cell* 84, 923–932.
- Petrova V & Russell C (2018) The evolution of seasonal influenza viruses. *Nat. Rev. Microbiol.* 16, 47–60.
- Polk DB & Peek RM, Jr (2010) *Helicobacter pylori*: gastric cancer and beyond. *Nat. Rev. Cancer* 10, 403–414.
- Ray K, Marteyn B, Sansonetti PJ & Tang CM (2009) Life on the inside: the intracellular lifestyle of cytosolic bacteria. *Nat. Rev. Microbiol.* 7, 333–340.
- Ribet D & Cossart P (2015) How bacterial pathogens colonize their hosts and invade deeper tissues. *Microbes Infect.* 17, 173–183.
- Sibley LD (2011) Invasion and intracellular survival by protozoan parasites. *Immunol. Rev.* 240, 72–91.
- Sun E, He J & Zhuang X (2013) Live cell imaging of viral entry. *Curr. Opin. Virol.* 3, 34–43.
- Tilney LG & Portnoy DA (1989) Actin filaments and the growth, movement, and spread of the intracellular bacterial parasite, *Listeria monocytogenes*. *J. Cell Biol.* 109, 1597–1608.
- Vink C, Rudenko G & Seifert HS (2012) Microbial antigenic variation mediated by homologous DNA recombination. *FEMS Microbiol. Rev.* 36, 917–948.
- Walsh D, Mathews MB & Mohr I (2013) Tinkering with translation: protein synthesis in virus-infected cells. *Cold Spring Harb. Perspect. Biol.* 5, a012351.
- Welch MD & Way M (2013) Arp2/3-mediated actin-based motility: a tail of pathogen abuse. *Cell Host Microbe* 14, 242–255.
- Worobey M, Watts TD, McKay RA . . . Jaffe HW (2016) 1970s and ‘Patient 0’ HIV-1 genomes illuminate early HIV/AIDS history in North America. *Nature* 39, 98–101.

### The Human Microbiota

- Fan Y & Pedersen O (2021) Gut microbiota in human metabolic health and disease. *Nat. Rev. Microbiol.* 19, 55–71.
- Fischbach MA (2019) Microbiome: focus on causation and mechanism. *Cell* 174, 785–790.
- Sender R, Fuchs S & Milo M (2016) Are we really vastly outnumbered? Revisiting the ratio of bacterial to host cells in humans. *Cell* 164, 337–340.